Online Bayesian Calibration under Gradual and Abrupt System Changes

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Abstract

Bayesian model calibration is central to digital twins and computer experiments, as it minimizes discrepancies between model outputs and field observations by optimizing calibration parameters and systematically identifying and correcting model bias. The classical Bayesian calibration framework relates computer model outputs to field observations by introducing latent parameters and a discrepancy function to capture bias, but suffers from identifiability issues due to confounding between parameters and discrepancy. In addition, it is formulated as an offline procedure under a stationary data-generating assumption, where observations arise from a fixed latent system. This assumption is often violated in digital twin applications, where physical systems evolve over time. Data assimilation methods enable sequential updates under gradual system changes, but they typically do not explicitly model systematic model bias and are not robust to abrupt regime shifts. In this paper, we first propose Bayesian Recursive Projected Calibration (BRPC), an online Bayesian calibration method that performs particle-based recursive updates of calibration parameters and Gaussian process modeling of discrepancy conditional on these parameters, enabling bias-aware adaptation under gradual system evolution while mitigating identifiability issues. We then extend BRPC by integrating an online change-point detection (CPD) module that identifies abrupt regime shifts and triggers resetting of the calibration process, yielding a unified framework for handling mixed nonstationarity (BRPC-CPD). We establish theoretical guarantees for both components, including error bounds for online calibration within regimes and performance guarantees for change-point detection. Empirical studies on synthetic and real datasets demonstrate that BRPC improves calibration tracking under gradual changes, while the integrated framework further enhances predictive accuracy and robustness under abrupt regime shifts compared to sliding-window Bayesian calibration and joint data-assimilation baselines.

1 Introduction

Bayesian calibration is a central problem in digital twins and computer model analysis. Computer simulators provide structured approximations to physical systems, typically involving unknown or artificial parameters that govern model behavior. These approximations are inherently imperfect, often leading to systematic discrepancies between model outputs and the true physical system. Consequently, computer models must be calibrated against observational data from the physical system to accurately estimate unknown model parameters and account for model bias. Estimating the model parameters and model biases is referred to as the model calibration problem.

The classical Bayesian calibration framework explicitly relates computer model outputs to field observations by introducing latent model parameters and a flexible discrepancy term to capture model bias [8]. A well-known limitation of this formulation is the identifiability issue arising from the confounding effects between model parameters and the discrepancy function. Projected calibration mitigates this issue by defining a calibration target prior to fitting a flexible discrepancy term, thereby

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improving identifiability [15, 18]. However, these classical formulations are inherently offline and assume a stationary data-generating process, in which observations arise from a fixed latent system. This assumption is frequently violated in modern applications characterized by nonstationarity.

Modern digital-twin applications, however, increasingly require calibration from streaming data, where observations arrive sequentially and operating conditions may evolve over time [16, 11]. In this online setting, the calibration target may drift, shift across regimes, or change together with the residual structure of the simulator. Online Bayesian calibration methods that preserve the identifiability logic of projected calibration remain comparatively less developed.

A straightforward approach to address these issues is to perform Bayesian calibration periodically using a fixed update interval and an associated sliding data window, which we refer to as sliding-window Bayesian calibration. However, Bayesian calibration is fundamentally an offline procedure, and frequent recalibration at short intervals can be computationally expensive. Increasing the update interval alleviates computational burden but may lead to substantial model bias when the system evolves rapidly. Filtering and data assimilation methods provide natural online tracking mechanisms [3, 4, 16], but they are not specifically designed to address the parameter–discrepancy identifiability issue central to Bayesian calibration, nor do they explicitly model systematic simulator bias. Moreover, while data assimilation can adapt to slow and gradual changes, it is generally not well suited for handling fast and abrupt regime shifts. Change-point and concept-drift methods offer tools for detecting nonstationarity [12, 1, 5], but they do not, by themselves, specify how the within-regime Bayesian calibration state should be represented or updated in the presence of simulator discrepancy.

An ideal model calibration approach should satisfy three key requirements: (1) an online calibration framework that incrementally updates both calibration parameters and discrepancy using streaming data, enabling adaptation to both gradual and abrupt system changes; (2) explicit modeling of systematic model bias, calibration parameters, and associated uncertainty; and (3) a restart mechanism that reinitializes the calibration process following abrupt regime changes. Such a mechanism is essential because data accumulated within a local regime can improve stability and predictive accuracy, but may introduce substantial bias once the underlying system undergoes a structural shift.

Toward this ideal, we propose Bayesian Recursive Projected Calibration (BRPC), an online projected-calibration framework for nonstationary data streams. BRPC explicitly relates computer model outputs to streaming observations through time-varying latent calibration parameters and a time-varying discrepancy term that captures systematic model bias. The method adopts a two-stage filtering strategy: it first updates the latent calibration parameters using incoming data to align the simulator with evolving system dynamics, and then updates a Gaussian process–based discrepancy model to capture residual bias conditional on the calibrated parameters. This two-stage filtering enforces a clear parameter–discrepancy separation, thereby improving identifiability of the two confounded components: calibration parameters are identified through simulator fit rather than residual absorption, while the discrepancy is subsequently learned for predictive correction.

BRPC is integrated with a restart mechanism that detects abrupt changes in data streams and reinitializes the online calibration process by resetting previously accumulated filtering states and minimizing the increase of biases due to before-change data. The restart decision is based on pre-update predictive evidence, which depends in part on the current recursive discrepancy state. Consequently, the quality of restart decisions depends not only on the change-point detection procedure, but also on the within-regime learner that generates the predictive evidence. A highly concentrated discrepancy posterior may cause routine within-regime variation to appear anomalous, while an overly adaptive discrepancy model may absorb emerging regime mismatches before the detector can identify a clear change.

We further analyze the interaction between online discrepancy learning and the restart mechanism, which leads to two complementary deployment strategies. The first modifies the restart rule to monitor shifts in predictive scores, while the second adjusts the discrepancy state to provide more segment-local residual evidence. We support the proposed framework with theoretical analysis of the recursive updates and the resulting predictive evidence used for control, and evaluate it on both controlled synthetic data streams and a Plant Simulation digital twin benchmark. Across a range of nonstationary settings, BRPC consistently improves calibration tracking and predictive accuracy compared to sliding-window Bayesian calibration and joint data-assimilation baselines.

2 Related Work

Static Bayesian calibration. Bayesian calibration with model discrepancy is commonly formulated by augmenting a simulator with a flexible bias term, often modeled as a Gaussian-process discrepancy or its variants [8, 17, 7, 20]. This improves predictive correction but creates parameter–discrepancy confounding: calibration parameters and discrepancy can explain the same simulator–reality mismatch. Projected calibration and Bayesian projected calibration address this problem by defining the calibration target before learning discrepancy [15, 18]. BRPC instead carries the projected-calibration principle into the online setting: the learned discrepancy does not enter the parameter update, and discrepancy is learned conditionally after the simulator anchor is set.

Sequential calibration, filtering, and data assimilation. Sequential Monte Carlo, ensemble Kalman filtering, and related data-assimilation methods provide natural tools for online state or parameter tracking [3, 4, 10, 13, 9]. In the digital-twin calibration literature, continuous calibration has also been studied by comparing particle-filter updates with sliding-window Kennedy–O'Hagan-style Bayesian calibration in an online setting [16]. These methods are effective local tracking mechanisms, especially under smooth evolution. Their update channel, however, is typically joint: parameter movement and model mismatch are absorbed through the same sequential assimilation mechanism. BRPC differs by enforcing a projected parameter channel and a conditional discrepancy channel, which makes the interaction with reset control explicit and analyzable.

Change-point detection and restart control. Change-point methods provide natural tools for handling abrupt nonstationarity. CUSUM changes through accumulated score or likelihood-ratio evidence [12, 19], while Bayesian online changepoint detection (BOCPD) maintains posterior uncertainty over run lengths and yields a soft mixture over candidate segment hypotheses [1, 14]. Restarted BOCPD further turns changepoint evidence into explicit state reset [2]. In online calibration, these detectors must be coupled with an adaptive within-regime learner, and the evidence monitored by the detector is itself produced by the current calibration state. BRPC studies this coupling in the presence of simulator discrepancy and projected recursive learning, where restart decisions and discrepancy adaptation can interact in nontrivial ways.

3 Problem Setup

We consider online calibration from streaming batches $B_t = \{(x_{t,k}, y_{t,k})\}_{k=1}^{K_t}$, with inputs X_t and responses Y_t. Let $y_s(x, \theta)$ denote the simulator response. The physical response may differ from the simulator because of both parameter mismatch and model discrepancy. We write the local observation model as

$$y_t(x) = \zeta_t(x) + \epsilon_t(x) = y_s(x, \theta_t) + \delta_t(x) + \epsilon_t(x), \qquad \epsilon_t(x) \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

where θ_t denotes the local calibration target and δ_t denotes the local discrepancy correction.

The goal is to track the evolving calibration target while preserving a meaningful separation between parameter learning and discrepancy learning. This separation is important because a flexible discrepancy can improve prediction but can also explain away the same simulator–reality mismatch that should inform calibration. In the projected-calibration view, the parameter target is therefore determined before fitting the flexible discrepancy; the discrepancy is then learned conditionally for predictive correction.

We focus on nonstationary streams. We use *drift* to denote gradual within-regime evolution of the local calibration target or residual structure. Under drift, consecutive batches remain close enough that information accumulated from recent past batches can still help stabilize tracking. We use *regime change* to denote an abrupt or structural change in the local calibration problem, such as a shift in operating mode, demand pattern, physical mechanism, or simulator–reality relationship. After such a change, information from the previous regime can become biased for the new regime even if it was useful before the change. Appendix F.1 gives a more formal version of this distinction.

This distinction motivates a local learning problem with optional restart. Within a regime, the learner updates the projected calibration and discrepancy states; after an operating-condition change, the local state may need to be reinitialized. The complication is that restart decisions use pre-update predictive evidence generated by the current learner, so learning and restart are coupled. The algorithmic ordering is summarized in Appendix A.

4 Bayesian Recursive Projected Calibration

4.1 From Bayesian projected calibration to online recursion

L_2-Projected calibration defines the calibration target before learning a flexible discrepancy. Let $\zeta(x)$ denote the physical response and let $y_s(x, \theta)$ denote the simulator. For any physical-response trajectory ζ, define its projected calibration parameter by

$$\Pi(\zeta) \in \arg\min_{\theta \in \Theta} \int_{\Omega} \{\zeta(x) - y_s(x, \theta)\}^2 \, dF_X(x), \tag{2}$$

where F_X is the design distribution. The discrepancy is learned only after this projection step, so a flexible discrepancy cannot change the calibration estimand.

Bayesian projected calibration applies this idea distributionally. In a typical static implementation, one first fits a probabilistic model for the physical response, such as a Gaussian-process posterior, and draws trajectories $\zeta^{(i)}$. Each trajectory induces a projected parameter $\theta^{(i)} = \Pi(\zeta^{(i)})$, and then a residual trajectory $r^{(i)}(x) = \zeta^{(i)}(x) - y_s(x, \theta^{(i)})$.

Conditional on this projected anchor, the discrepancy posterior is the usual Gaussian-process update. Equivalently, for a GP prior $p_0(\delta)$ and Gaussian residual likelihood,

$$q^{\delta,(i)} = \arg\min_{q} \left\{ -\mathbb{E}_q \left[\log p(r^{(i)} \mid \delta) \right] + \mathrm{KL}(q(\delta) \,\|\, p_0(\delta)) \right\}. \tag{3}$$

Thus static BPC is a two-stage Bayesian procedure: posterior uncertainty in the physical response induces uncertainty in the projected calibration parameter, and discrepancy is then learned conditionally on each projected parameter.

BRPC turns this static two-stage procedure into an online recursion. Directly repeating static BPC at every batch would require repeatedly refitting the physical-response posterior, sampling new trajectories, projecting each trajectory, and refitting discrepancy from anchor-dependent residuals. This is both computationally expensive and poorly matched to nonstationary streams, where old batches should not always be treated as exchangeable with the current batch. BRPC instead maintains recursive posterior states for the projected parameter and the conditional discrepancy. Each new batch updates these states through KL-regularized online steps, giving an online analogue of the two stages above.

4.2 Projected particle update

Let q^{θ}_{t-1} denote the current posterior over the local projected calibration parameter. Given a local evolution model $p(\theta_t \mid \theta_{t-1})$, define the predictive parameter prior

$$\bar{q}^{\theta}_{t|t-1}(\theta_t) = \int p(\theta_t \mid \theta_{t-1}) q^{\theta}_{t-1}(\theta_{t-1}) \, d\theta_{t-1}. \tag{4}$$

BRPC updates the parameter posterior using a discrepancy-free projected likelihood

$$p_{\mathrm{proj}}(Y_t \mid X_t, \theta_t) = \mathcal{N}\left(Y_t; y_s(X_t, \theta_t), \Sigma_{\theta,t}\right). \tag{5}$$

The learned discrepancy does not enter this parameter-weighting step.

Proposition 1 (Projected parameter update)**.** *The KL-regularized projected update*

$$q^{\theta}_t = \arg\min_{q} \left\{ -\eta_\theta \mathbb{E}_q \left[\log p_{\mathrm{proj}}(Y_t \mid X_t, \theta_t) \right] + \mathrm{KL}\Big(q(\theta_t) \,\|\, \bar{q}^{\theta}_{t|t-1}(\theta_t) \Big) \right\} \tag{6}$$

has solution

$$q^{\theta}_t(\theta_t) \propto \bar{q}^{\theta}_{t|t-1}(\theta_t) \, p_{\mathrm{proj}}(Y_t \mid X_t, \theta_t)^{\eta_\theta}. \tag{7}$$

If $\bar{q}^{\theta}_{t|t-1}$ is represented by particles $\{(\theta^{(i)}_{t|t-1}, w^{(i)}_{t|t-1})\}_{i=1}^N$, then (7) is approximated by

$$w^{(i)}_t \propto w^{(i)}_{t|t-1} p_{\mathrm{proj}}\Big(Y_t \mid X_t, \theta^{(i)}_{t|t-1}\Big)^{\eta_\theta}, \qquad \sum_{i=1}^{N} w^{(i)}_t = 1. \tag{8}$$

The corresponding particle implementation propagates particles through $p(\theta_t \mid \theta_{t-1})$, reweights them by $p_{\text{proj}}(Y_t \mid X_t, \theta_t)^{\eta_\theta}$, and resamples when the effective sample size is small. This particle filter is the online analogue of the pushforward uncertainty in static BPC: it maintains a distribution over the evolving projected calibration target. When $\Sigma_{\theta,t} = \sigma_t^2 I$, the negative log projected likelihood is proportional to the empirical squared projected loss on the current batch, so the update remains tied to the projected calibration criterion. Details are given in Appendix B.2.

4.3 Conditional recursive discrepancy update

After the projected parameter update, discrepancy is learned conditionally on the current parameter particles. For particle i, define the projected residual batch $r_t^{(i)} = Y_t - y_s(X_t, \theta_t^{(i)})$.

Let $q_{t,-}^{\delta,(i)}$ denote the Gaussian discrepancy prior available before assimilating the current residual batch for particle i:

$$q_{t,-}^{\delta,(i)}(u_t^{(i)}) = \mathcal{N}(a_t^{(i)}, P_t^{(i)}), \qquad M_t^{(i)} = (P_t^{(i)})^{-1}. \tag{9}$$

Here $u_t^{(i)}$ is the finite discrepancy state used by the chosen BRPC variant. For a fixed-support state, $q_{t,-}^{\delta,(i)}$ is simply the previous discrepancy posterior carried forward. For an expanding-support state, it is the previous posterior extended to the enlarged support by GP conditioning. The construction of $q_{t,-}^{\delta,(i)}$ for each BRPC variant is given in Appendix B.4. Let $G_t^{(i)}$ map this carried discrepancy state to the current inputs X_t. The residual likelihood is

$$r_t^{(i)} \mid u_t^{(i)} \sim \mathcal{N}\Big(G_t^{(i)} u_t^{(i)}, R_t\Big), \tag{10}$$

where R_t is the residual-likelihood covariance.

Proposition 2 (Recursive conditional discrepancy update)**.** *Under* (9) *and* (10)*, the KL-regularized discrepancy update for particle i,*

$$q_t^{\delta,(i)} = \arg\min_q \left\{ -\eta_\delta \mathbb{E}_q \left[\log p(r_t^{(i)} \mid G_t^{(i)} u_t^{(i)}) \right] + \mathrm{KL}\Big(q(u_t^{(i)}) \,\|\, q_{t,-}^{\delta,(i)}(u_t^{(i)}) \Big) \right\}, \tag{11}$$

has Gaussian solution $q_t^{\delta,(i)} = \mathcal{N}(m_t^{(i)}, C_t^{(i)})$ with

$$(C_t^{(i)})^{-1} = (P_t^{(i)})^{-1} + \eta_\delta (G_t^{(i)})^\top R_t^{-1} G_t^{(i)}, \qquad m_t^{(i)} = C_t^{(i)} \left\{ (P_t^{(i)})^{-1} a_t^{(i)} + \eta_\delta (G_t^{(i)})^\top R_t^{-1} r_t^{(i)} \right\}. \tag{12}$$

The particle-specific construction is the cleanest Bayesian analogue of static BPC, because each parameter particle induces its own residual batch and its own conditional discrepancy posterior. In the main experiments, we use a shared expert-level approximation for stability and computational efficiency. Instead of maintaining one discrepancy posterior per particle, we form the weighted residual batch $r_t = \sum_{i=1}^N w_t^{(i)} r_t^{(i)} = Y_t - \sum_{i=1}^N w_t^{(i)} y_s(X_t, \theta_t^{(i)})$, and update a single discrepancy posterior using the same formula as Proposition 2, with the particle index removed. This shared update reduces the number of adaptive discrepancy posteriors and stabilizes the residual signal in finite-particle implementations; The full particle-specific construction and the empirical ablation comparing it with the shared construction are given in Appendix B.5 and Appendix J.4.

4.4 Prediction and prequential likelihood

BRPC prediction combines projected parameter particles with conditional discrepancy posteriors. After assimilating batch t, each particle induces a Gaussian predictive law $p_t^{(i)}(Y_\star \mid X_\star) = \mathcal{N}(Y_\star; \mu_{\star,t}^{(i)}, \Sigma_{\star,t}^{(i)})$, where $\mu_{\star,t}^{(i)}$ and $\Sigma_{\star,t}^{(i)}$ include the simulator prediction, the conditional GP discrepancy mean, observation noise, and discrepancy uncertainty. The BRPC predictive distribution is the particle mixture

$$p_t(Y_\star \mid X_\star) = \sum_{i=1}^N w_t^{(i)} p_t^{(i)}(Y_\star \mid X_\star). \tag{13}$$

In prequential use, the same likelihood is evaluated before assimilating the incoming batch and provides the evidence for restart decisions. Full GP predictive maps for the particle-specific, fixed-support, and shared implementations are given in Appendix D.

5 BRPC under Regime Change: Restart and Compatibility

The recursive BRPC update is designed to be stable within a local regime. The projected particle filter uses a local evolution prior, and the discrepancy update is KL-regularized toward the pre-update discrepancy state. These features are useful under smooth drift: they allow the learner to reuse recent information rather than refitting from scratch at every batch. Under an abrupt regime change, however, the same transported state can become stale. Parameter particles and discrepancy information carried from the old regime may no longer cover the new local calibration problem, so the learner can adapt too slowly or remain biased by outdated calibration evidence.

This motivates coupling BRPC with a change-point-driven restart mechanism. The role of restart is not to replace the within-regime learner, but to decide when the current local calibration state should be discarded and a fresh local state should be initialized. Throughout this section, let $p_e^{\mathrm{pre}}(Y_t \mid X_t)$ denote the predictive law of expert e formed before assimilating batch t, and let

$$L_{e,t}^{\mathrm{pre}} := -\log p_e^{\mathrm{pre}}(Y_t \mid X_t)$$

be the corresponding prequential loss. The predictive formulas are given in Appendix D.

BOCPD as a natural first deployment. BOCPD is a natural first restart mechanism for BRPC. It is Bayesian, it can use the BRPC pre-update predictive likelihood directly, and it maintains multiple candidate segment hypotheses rather than committing to a single local trajectory. This multi-expert structure can help recover after a regime change by allowing a fresh expert to compete with continuation experts that carry stale parameter or discrepancy states.

In B-BRPC, BOCPD-style restart control is run on expert-conditional pre-update batch predictive densities. With restart hazard h_t, continuation branches receive mass proportional to $(1-h_t)\, w_{e,t-1}\, p_e^{\mathrm{pre}}(Y_t \mid X_t)$, while a fresh restart branch receives mass proportional to $h_t\, p_{\mathrm{restart}}^{\mathrm{pre}}(Y_t \mid X_t)$. Standard BOCPD maintains a soft posterior over run lengths; to study explicit state reset, we use the restarted BOCPD rule, which triggers a hard restart when the fresh branch overtakes the anchor expert launched at the most recent reset [1, 2]. The full forecaster recursion is given in Appendix A.1.

This deployment is attractive because it turns BRPC predictive evidence into segment-level restart decisions. Its success, however, depends on the compatibility between the predictive evidence produced by the recursive learner and the likelihood competition used by BOCPD.

Why compatibility can fail. The issue is structural rather than only a matter of tuning the BOCPD hazard. BOCPD compares continuation and restart hypotheses through pre-update predictive likelihoods, but these likelihoods are shaped by the carried BRPC discrepancy state. Recursive discrepancy learning can therefore distort the evidence used for run-length competition.

There are two opposite failure modes. First, the continuation expert can become overconfident. As the recursive discrepancy posterior contracts, its predictive covariance can become too small; then ordinary within-regime mismatch produces a large surprise, and BOCPD may interpret this as evidence for restart. This increases false alarms. Second, the discrepancy learner can be too adaptive. If the active discrepancy state absorbs emerging mismatch quickly, the predictive contrast between stale and fresh experts is blurred, and BOCPD may no longer see a clean regime-change signal. Thus the same recursive learner can make BOCPD restart too often in one regime and restart too weakly in another. This motivates treating restart design as part of the BRPC deployment rather than as a modular detector attached afterward. We study two complementary responses.

Controller-side response: score-based restart. The first response keeps the recursive BRPC learner but changes the restart rule. Instead of comparing many segment experts by full predictive likelihoods, C-BRPC monitors the current learner's pre-update score stream. Define

$$s_t := -\frac{1}{K_t} \log p^{\mathrm{pre}}(Y_t \mid X_t), \tag{14}$$

where K_t is the batch size. We standardize the score within the current segment, for example by $s_t^{\mathrm{std}} = (s_t - \hat{\mu}_{t-1})/\max(\hat{\sigma}_{t-1}, \sigma_{\min})$, where $\hat{\mu}_{t-1}$ and $\hat{\sigma}_{t-1}$ are running estimates from previous pre-update scores. The window-limited CUSUM statistic is

$$G_t = \max_{1 \le m \le W} \sqrt{m}\, \left(\bar{s}_{t-m+1:t}^{\mathrm{std}} - \kappa\right)_+, \qquad \bar{s}_{t-m+1:t}^{\mathrm{std}} = \frac{1}{m} \sum_{u=t-m+1}^{t} s_u^{\mathrm{std}}. \tag{15}$$

A restart is triggered when $G_t > h$. This changes the question from "which segment expert best explains the batch?" to "has the current predictive score stream shifted enough to justify restart?" The resulting controller is less dependent on calibrated likelihood competition across many adaptive experts.

State-side response: residual re-anchoring. The second response keeps BOCPD-style restart control but changes the discrepancy state exposed to BOCPD. In residual re-anchoring, each active segment rebuilds its discrepancy dataset relative to the current simulator anchor and refits the segment GP. For expert e at time t, let $\bar{\mu}_{e,t}(x)$ denote the current simulator anchor. The re-anchored residual dataset is

$$\mathcal{D}_{e,t}^{\text{res}} = \left\{ \left(x_{u,k},\, y_{u,k} - \bar{\mu}_{e,t}(x_{u,k}) \right) \right\}_{(u,k)\in\mathcal{I}_{e,t}},$$

where $\mathcal{I}_{e,t}$ indexes the observations assigned to the current segment. The discrepancy posterior is then refit on $\mathcal{D}_{e,t}^{\text{res}}$; details are given in Appendix C. This is computationally heavier than the recursive update, but it produces more segment-local residual evidence and removes the direct path by which stale recursive discrepancy state is transported into the BOCPD likelihood competition.

The two responses diagnose different aspects of the same compatibility problem. Score-based restart changes the controller so that it monitors instability in a single score stream, while residual re-anchoring changes the discrepancy state so that BOCPD-facing evidence is more segment-local. In the experiments, score-based restart serves as the more robust default across mixed nonstationarity, whereas residual re-anchoring is most effective when abrupt jumps dominate and segment-local refitting is affordable.

6 Theory Overview

The theory supports three aspects of the BRPC design. Theorem 1 shows that the recursive discrepancy update is a principled within-regime tracker. Proposition 3 identifies how recursive discrepancy learning can distort BOCPD restart evidence, and Proposition 4 gives idealized false-alarm and detection guarantees for the score-based controller.

Write the recursive discrepancy update in mean form as

$$z_t = \arg\min_z \left\{ \eta_\delta \ell_t(z) + \frac{1}{2}\|z - A_t z_{t-1}\|_{M_t}^2 \right\}, \qquad \ell_t(z) = \frac{1}{2}\|r_t - G_t z\|_{R_t^{-1}}^2, \tag{16}$$

where z_t is the BRPC discrepancy posterior mean, A_t carries the previous discrepancy state to the current state representation, and M_t is the pre-update precision.

Assumption 1 (Stable state transport)**.** *There exists $\gamma \in [0,1)$ such that*

$$A_t^\top M_t A_t \preceq \gamma M_{t-1} \qquad \textit{for all } t \geq 1. \tag{17}$$

This is an algorithmic stability condition on carried discrepancy information; sufficient conditions based on process noise, nugget regularization, and fixed-support approximations are given in Appendix G.7.

Theorem 1 (Recursive BRPC tracking)**.** *Under Assumption 1, for any reference discrepancy sequence $\{v_t\}_{t=1}^T$ in the same carried state space as z_t,*

$$\sum_{t=1}^{T} [\ell_t(z_t) - \ell_t(v_t)] \leq \frac{1}{2\eta_\delta}\|v_0 - z_0\|_{M_0}^2 + \frac{1}{2\eta_\delta(1-\gamma)} \sum_{t=1}^{T} \|v_t - A_t v_{t-1}\|_{M_t}^2. \tag{18}$$

The reference sequence v_t is not observed data, but an oracle discrepancy state one would like to track. In fixed support, $A_t = I$ and the variation term is ordinary path variation; in expanding support, A_t transports the reference state across supports. This variation is small under smooth drift and large under abrupt jumps, supporting the division of labor between recursive BRPC updates and restart. The proof is given in Appendix E.2.

For restart evidence, recall that each BRPC expert induces a pre-update predictive law before the incoming batch is assimilated. To isolate the mechanism, consider a Gaussian approximation to the continuation and fresh expert predictive laws, $p_c(y) = \mathcal{N}(y; \mu_c, \Sigma_c)$, and $p_n(y) = \mathcal{N}(y; \mu_n, \Sigma_n)$, with restart hazard $h_t \in (0,1)$.

Proposition 3 (Gaussian restart odds)**.** *If the incoming batch is generated from $Y_t \sim \mathcal{N}(\mu_\star, \Sigma_\star)$, then the expected one-step log posterior odds in favor of restart satisfy*

$$\mathbb{E}\left[\log \frac{w_t^{\text{restart}}}{w_t^{\text{cont}}}\right] = \log \frac{h_t}{1-h_t} + \frac{1}{2}\left\{ \log \frac{\det \Sigma_c}{\det \Sigma_n} + \operatorname{tr}\left[(\Sigma_c^{-1} - \Sigma_n^{-1})\Sigma_\star\right] + \|\mu_\star - \mu_c\|^2_{\Sigma_c^{-1}} - \|\mu_\star - \mu_n\|^2_{\Sigma_n^{-1}} \right\}. \tag{19}$$

This expression gives an evidence-drift view of controller–learner compatibility. Within a stable regime, the expected log posterior odds for restart should remain negative, or below the hard-restart margin, so that ordinary residual fluctuation does not accumulate into false restart evidence. After a true regime change, the expected odds should become positive before the continuing discrepancy state absorbs the mismatch. A too-small continuation covariance or biased continuation mean promotes false alarms through the Σ_c^{-1}-weighted covariance and bias terms, whereas overly fast discrepancy adaptation reduces the fresh expert's likelihood advantage and delays restart. Thus compatibility is governed by the balance between continuation sharpness, continuation bias, and recursive adaptation speed. The proof is given in Appendix G.3.

Finally, let s_t be the batch-normalized pre-update score and s_t^{std} its standardized version. Define

$$G_t = \max_{1 \le m \le W} \sqrt{m}\left(\bar{s}^{\text{std}}_{t-m+1:t} - \kappa\right)_+ .$$

Proposition 4 (Idealized wCUSUM false-alarm and detection bounds)**.** *Suppose that, under no change, standardized score averages are sub-Gaussian in the sense that*

$$\mathbb{P}\big(\sqrt{m}\,\bar{s}^{\text{std}}_{t-m+1:t} > h + \sqrt{m}\,\kappa\big) \le \exp\{-(h + \sqrt{m}\,\kappa)^2/2\}$$

for all t and $1 \le m \le W$. Then

$$\mathbb{P}\left(\max_{1 \le t \le T} G_t > h\right) \le TWe^{-h^2/2}.$$

If after a regime change the standardized score mean is persistently shifted by at least $\delta > \kappa$, then detection occurs within a window of order $h^2/(\delta - \kappa)^2$ up to logarithmic factors.

The concentration assumption abstracts the score standardization step; it is intended as an idealized sub-Gaussian model for within-regime prequential scores, with empirical sensitivity checks reported in Appendix J.7. The proof, including the separate false-alarm and detection statements, is given in Appendix G.4.

7 Experiments

We evaluate BRPC on two benchmark families: a synthetic streaming benchmark and a plant-simulation benchmark. The synthetic benchmark is designed to isolate local adaptation, restart quality, and controller–learner mismatch under nonstationarity. The plant-simulation benchmark tests whether the same qualitative patterns persist in a less stylized digital-twin setting.

The synthetic benchmark is adapted from the static calibration example of Gu and Wang [6]. The simulator is $y_s(x, \theta) = \sin(\theta x) + 5x$, and the physical response is based on $\zeta_t(x) = 5x\cos(\omega_t x/2) + 5x$, observed with Gaussian noise. In the original static example, $\omega_t = 15$ gives a fixed projected calibration target; we turn this example into an online benchmark by allowing ω_t and the corresponding projected target to drift smoothly, jump abruptly, or combine both behaviors. We study drifting, sudden-change, and mixed scenario families, with annotated abrupt changes in the latter two for event-level restart evaluation. Detailed benchmark definitions are given in Appendix I.

We compare BOCPD-based restart control (B-BRPC), score-based wCUSUM restart control (C-BRPC), and BOCPD with residual re-anchoring (B-BRPC-RRA). In the main tables, BRPC denotes the default recursive implementation used throughout the main text; implementation-level variants such as expanding-support, fixed-support, and proxy-memory versions are reported in Appendix J. We also include BC(80), a sliding-window Kennedy–O'Hagan recalibration baseline that refits the static Bayesian calibration model using the most recent 80 observations, and DA, a filtering-style

baseline adapted from the continuous-calibration comparison of Ward et al. [16]. Our DA baseline augments the filtering update with random-walk parameter evolution.

Results average 25 synthetic seeds and 10 plant seeds, with scalar metrics reported as mean $\pm$ standard error. Precision@2 and Recall@2 use a two-batch tolerance window around annotated abrupt jumps; in mixed settings, drift-triggered restarts may therefore reduce precision even when useful for local adaptation. Relative y-RMSE on the plant benchmark is normalized by the mode-specific response scale. Delay, runtime, BOCPD-controlled filtering diagnostics, and full ablations are deferred to Appendix J.

Table 1: Main benchmark results on the synthetic and plant-simulation scenario families. For plant-simulation, the response metric is relative y-RMSE. Arrows indicate preferred directions for performance metrics; restart count is a controller diagnostic.

Scenario / Mode	Method	θ-RMSE ↓	Response RMSE ↓	Restarts	Precision@2 ↑	Recall@2 ↑
Synthetic benchmark						
Drifting	BC(80)	0.142±0.010	0.622±0.242	–	–	–
	DA	0.090±0.030	1.909±0.116	–	–	–
	B-BRPC	0.014±0.002	0.484±0.225	11.774±1.733	–	–
	C-BRPC	0.015±0.002	0.580±0.262	4.774±0.419	–	–
	B-BRPC-RRA	0.015±0.002	0.437±0.115	8.128±2.345	–	–
Sudden(3)	BC(80)	0.166±0.079	0.839±0.437	–	–	–
	DA	0.178±0.162	1.994±0.070	–	–	–
	B-BRPC	0.026±0.023	0.607±0.285	10.287±4.158	0.314	0.926
	C-BRPC	0.027±0.024	0.671±0.321	3.706±1.209	0.725	0.834
	B-BRPC-RRA	0.018±0.012	0.512±0.201	3.050±0.218	0.986	0.999
Mixed(3)	BC(80)	0.125±0.020	0.647±0.103	–	–	–
	DA	0.092±0.039	1.873±0.148	–	–	–
	B-BRPC	0.021±0.016	0.464±0.070	11.865±2.675	0.156	0.960
	C-BRPC	0.020±0.014	0.535±0.096	3.493±1.005	0.392	0.720
	B-BRPC-RRA	0.021±0.016	0.505±0.066	4.813±1.565	0.409	0.960
Plant-simulation benchmark						
Drifting	BC(80)	3.936±0.123	3.659±0.045	–	–	–
	DA	9.562±0.276	5.027±0.189	–	–	–
	B-BRPC	0.800±0.018	0.988±0.031	15.7±0.082	–	–
	C-BRPC	0.809±0.024	1.001±0.027	7.0±0.000	–	–
	B-BRPC-RRA	0.788±0.022	1.109±0.042	17.9±0.320	–	–
Sudden(5)	BC(80)	6.023±0.250	1.659±0.147	–	–	–
	DA	4.259±0.238	1.000±0.153	–	–	–
	B-BRPC	0.957±0.086	0.164±0.027	7.4±0.970	0.516	0.760
	C-BRPC	1.056±0.238	0.169±0.035	4.3±0.670	0.762	0.660
	B-BRPC-RRA	1.048±0.411	0.128±0.030	5.0±0.000	1.000	1.000
Mixed(≈2–3)	BC(80)	4.502±1.123	3.034±0.345	–	–	–
	DA	9.669±1.276	5.646±0.459	–	–	–
	B-BRPC	1.580±0.898	0.554±0.214	12.4±5.276	0.184	0.950
	C-BRPC	1.582±1.111	0.561±0.178	5.7±2.500	0.312	0.747
	B-BRPC-RRA	1.537±0.857	0.569±0.176	13.0±5.330	0.186	1.000

Main benchmark results. Table 1 shows that BRPC improves projected-parameter tracking over BC(80) and DA across both synthetic and plant-simulation benchmarks. B-BRPC and C-BRPC have similar θ-RMSE, so their main difference is restart behavior: B-BRPC tends to achieve high recall with excessive restarts, while C-BRPC reduces restart burden and improves event-level precision. B-BRPC-RRA is strongest in jump-dominated settings (*Sudden(3)* and *Sudden(5)*), where segment-local residual refitting gives near-ideal restart behavior. Thus C-BRPC is the safer default under heterogeneous nonstationarity, whereas B-BRPC-RRA is most useful when abrupt segmentation dominates and refitting is affordable. Additional ablations and the high-dimensional diagnostic are reported in Appendices J.4 and J.9.

8 Conclusion

We introduced Bayesian Recursive Projected Calibration (BRPC), an online Bayesian calibration framework for streaming data under simulator mismatch and nonstationarity. BRPC extends the identifiability logic of Bayesian projected calibration by separating a discrepancy-free projected particle update for calibration parameters from a conditional recursive Gaussian-process update for discrepancy. This allows online updating while keeping discrepancy from directly altering the projected calibration target.

We also showed that change-point-driven restart must be designed together with the recursive learner, because discrepancy learning shapes the pre-update predictive evidence used by restart mechanisms. Across synthetic and plant-simulation benchmarks, score-based restart is the more robust default under heterogeneous nonstationarity, while BOCPD with residual re-anchoring is most useful in jump-dominated regimes when segment-local refitting is affordable. Future work should test broader high-dimensional and real digital-twin settings and develop restart mechanisms with stronger false-alarm control or softer memory-reset strategies.

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Appendix organization. Appendix A gives the algorithmic ordering and restart-controller pseudocode. Appendix B gives BRPC construction details, including the static BPC template, the projected parameter update, discrepancy state maps, and shared versus particle-specific residual constructions. Appendix C gives the residual re-anchoring construction. Appendix D gives predictive laws. Appendix E proves the BRPC tracking result and gives a sufficient condition for stable state transport. Appendix F states auxiliary theory for drift, BOCPD restart, identifiability, and residual-data selection, with complete proofs and additional contraction and contamination identities in Appendix G. Appendix H describes the synthetic and plant-simulation benchmarks. Appendix I reports additional ablations, sensitivity checks, trajectory diagnostics, and the high-dimensional physical-projected diagnostic.

A Algorithms

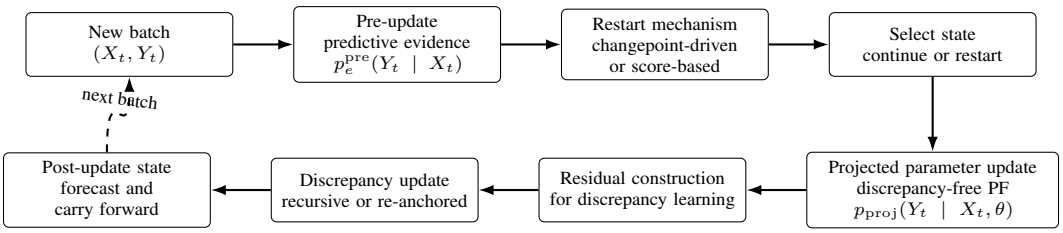


Figure 1: One-step BRPC update with restart control. The controller acts on pre-update predictive evidence before the selected local learner assimilates the current batch. After the projected parameter update, residuals are constructed relative to the selected calibration state and used for discrepancy learning; the precise residual construction depends on the BRPC variant.

Algorithm 1 Projected particle update

1: Propagate particles: $\theta_{t|t-1}^{(i)} \sim p(\theta_t \mid \theta_{t-1}^{(i)})$.
2: Compute projected likelihoods: $\ell_t^{(i)} = p_{\mathrm{proj}}(Y_t \mid X_t, \theta_{t|t-1}^{(i)})$.
3: Reweight: $w_t^{(i)} \propto w_{t|t-1}^{(i)} (\ell_t^{(i)})^{\eta_\theta}$.
4: Normalize weights and resample if the effective sample size is below a threshold.

Algorithm 2 B-BRPC: BRPC with Restart-BOCPD control

Require: BRPC variant $\in \{\mathrm{E}, \mathrm{P}, \mathrm{F}, \mathrm{RRA}\}$, number of particles N, ESS threshold τ_{ESS}, restart hazard h_t, previous expert set $\mathcal{E}_{t-1}$.

for $t = 1, 2, \ldots$ **do**
  Observe batch (X_t, Y_t).
  for each continuation expert $e \in \mathcal{E}_{t-1}$ **do**
    Form pre-update predictive law $p_e^{\mathrm{pre}}(Y_t \mid X_t)$.
  end for
  Form the fresh-branch pre-update predictive law $p_{\mathrm{new}}^{\mathrm{pre}}(Y_t \mid X_t)$ from the restart prior.
  Run the Restart-BOCPD weight update using $\{p_e^{\mathrm{pre}}(Y_t \mid X_t)\}_{e \in \mathcal{E}_{t-1}}$, $p_{\mathrm{new}}^{\mathrm{pre}}(Y_t \mid X_t)$, and hazard h_t.
  Apply the hard-restart and pruning rules to obtain the candidate expert set $\mathcal{E}_t^{\mathrm{cand}}$.
  for each expert $e \in \mathcal{E}_t^{\mathrm{cand}}$ **do**
    if e is fresh **then**
      Initialize parameter and discrepancy states from the restart priors.
    else
      Carry forward the discrepancy state.
    end if
    Update parameter particles using Algorithm 1.
    Construct the residual object required by the chosen BRPC variant.
    Update the discrepancy state using Proposition 2.
  end for
  Set $\mathcal{E}_t \leftarrow \mathcal{E}_t^{\mathrm{cand}}$.
end for

Algorithm 3 C-BRPC: BRPC with wCUSUM control

Require: BRPC variant $\in \{\mathrm{E}, \mathrm{P}, \mathrm{F}\}$, number of particles N, ESS threshold τ_{ESS}, window limit W, drift allowance κ, threshold h, active expert state.

for $t = 1, 2, \ldots$ **do**
  Observe batch (X_t, Y_t).
  Form the active expert's pre-update predictive law $p^{\mathrm{pre}}(Y_t \mid X_t)$.
  Compute the batch-normalized pre-update score

$$s_t = -\frac{1}{K_t} \log p^{\mathrm{pre}}(Y_t \mid X_t).$$

  Standardize s_t using previous scores in the current segment and update

$$G_t = \max_{1 \le m \le W} \sqrt{m} \left(\bar{s}_{t-m+1:t}^{\mathrm{std}} - \kappa \right)_+ .$$

  if $G_t > h$ **then**
    Reset: initialize parameter and discrepancy states from the restart priors.
    Reset the score-standardization history.
  end if
  Update parameter particles using Algorithm 1.
  Construct the residual object required by the chosen BRPC variant.
  Update the discrepancy state using Proposition 2.
end for

A.1 BOCPD-style forecaster recursion with hard restart

BOCPD maintains a posterior mixture over possible run lengths by updating each candidate segment expert with its pre-update predictive likelihood. In our setting, a run-length expert is a BRPC local learner, and the one-step likelihood is replaced by the batch pre-update predictive evidence $p_e^{\mathrm{pre}}(Y_t \mid X_t)$. Following the hard-restart principle of restarted BOCPD, we use this BOCPD forecaster recursion to compare continuation experts with a newly initialized expert, and declare

a restart when a post-anchor candidate overtakes the anchor expert. The only extension from the pointwise setting is that all likelihood terms below are batch predictive likelihoods.

Let $\mathcal{E}_{t-1}$ denote the active expert set before observing batch t, with controller weights $\{w_{e,t-1}\}_{e\in\mathcal{E}_{t-1}}$. The corresponding mixture predictive law is

$$p^{\mathrm{mix}}(Y_t \mid X_t, \mathcal{F}_{t-1}) = \sum_{e\in\mathcal{E}_{t-1}} w_{e,t-1}\, p_e^{\mathrm{pre}}(Y_t \mid X_t). \tag{20}$$

With restart hazard $h_t \in (0,1)$, continuation experts and the fresh restart branch receive unnormalized weights

$$\widetilde{w}_{e,t} = \begin{cases} (1-h_t)\, w_{e,t-1}\, p_e^{\mathrm{pre}}(Y_t \mid X_t), & e \in \mathcal{E}_{t-1}, \\ h_t\, p_{\mathrm{new}}^{\mathrm{pre}}(Y_t \mid X_t), & e = e_t^{\mathrm{new}}, \end{cases} \qquad w_{e,t} = \frac{\widetilde{w}_{e,t}}{\sum_{e'} \widetilde{w}_{e',t}}. \tag{21}$$

Here $p_{\mathrm{new}}^{\mathrm{pre}}(Y_t \mid X_t)$ is the batch predictive law of a freshly initialized BRPC expert.

Let e_t^{anc} denote the anchor expert launched at the most recent hard restart. For each expert $e \in \mathcal{E}_t$, let s_e denote the segment start time represented by that expert. Following the restarted-BOCPD hard decision rule, we declare a restart when some post-anchor candidate has larger posterior mass than the anchor:

$$\mathrm{Restart}_t = \mathbf{1}\left\{ \max_{e\in\mathcal{E}_t:\, s_e > s_{e_t^{\mathrm{anc}}}} w_{e,t} > w_{e_t^{\mathrm{anc}},t} \right\}. \tag{22}$$

When $\mathrm{Restart}_t = 1$, the selected post-anchor candidate becomes the new anchor and the local BRPC state is reinitialized for the new segment.

We use the hard-restart decision in the main experiments because it makes restart quality directly measurable through event-level metrics such as Precision@2, Recall@2, and Delay@2. This choice is not meant to rule out the standard soft BOCPD deployment: in applications where predictive averaging is preferred over explicit state reset, one can retain the BOCPD posterior mixture over BRPC experts and use the mixture predictive law in (20) without committing to a single restarted state.

B BRPC Construction Details

This appendix gives the details suppressed from Section 4. We first recall the static BPC construction, then prove the two closed-form BRPC updates used in the main text, and finally describe the state maps for BRPC-E, BRPC-P, BRPC-F, and the residual constructions.

B.1 Static Bayesian projected calibration

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ denote a static calibration dataset. Bayesian projected calibration first places a posterior distribution on the physical response ζ, denoted by $Q^\zeta(d\zeta \mid \mathcal{D})$. For each posterior draw $\zeta^{(m)}$, the projected calibration parameter is

$$\theta^{(m)} = \Pi(\zeta^{(m)}) \in \operatorname*{arg\,min}_{\theta\in\Theta} \int \{\zeta^{(m)}(x) - y_s(x,\theta)\}^2\, dF_X(x).$$

The induced projected calibration posterior is the pushforward The collection of projected parameters

$$\theta^{(m)} = \Pi(\zeta^{(m)}), \qquad \zeta^{(m)} \sim Q^\zeta(\cdot \mid \mathcal{D}),$$

defines the Bayesian projected calibration posterior over θ. This distributional step is the Bayesian part of BPC: uncertainty in the physical response induces uncertainty in the projected calibration parameter.

Conditional on a projected anchor $\theta^{(m)}$, discrepancy is learned from residuals

$$r^{(m)}(X) = \zeta^{(m)}(X) - y_s(X, \theta^{(m)}).$$

If $\delta \sim GP(0, k_\phi)$ and the residual likelihood is

$$r^{(m)}(X) \mid \delta \sim \mathcal{N}(\delta(X), R),$$

then the conditional discrepancy posterior is

$$q_m^\delta(\delta) \propto p(r^{(m)}(X) \mid \delta)\, p_0(\delta).$$

Equivalently,

$$q_m^\delta = \arg\min_q \left\{ -\mathbb{E}_q[\log p(r^{(m)}(X) \mid \delta)] + \mathrm{KL}(q(\delta) \,\|\, p_0(\delta)) \right\}.$$

For any finite support S, this posterior is Gaussian by standard GP conditioning. This variational form is the static template that BRPC converts into a recursive update.

B.2 Projected parameter update

Proof of Proposition 1. The objective in (6) is

$$\mathcal{J}(q) = -\eta_\theta \int q(\theta_t) \log p_{\mathrm{proj}}(Y_t \mid X_t, \theta_t)\, d\theta_t + \int q(\theta_t) \log \frac{q(\theta_t)}{\bar{q}^\theta_{t|t-1}(\theta_t)}\, d\theta_t.$$

Using a Lagrange multiplier for the constraint $\int q(\theta_t) d\theta_t = 1$, the first-order condition gives

$$-\eta_\theta \log p_{\mathrm{proj}}(Y_t \mid X_t, \theta_t) + \log q(\theta_t) - \log \bar{q}^\theta_{t|t-1}(\theta_t) + 1 + \lambda = 0.$$

Hence

$$q^\theta_t(\theta_t) \propto \bar{q}^\theta_{t|t-1}(\theta_t) p_{\mathrm{proj}}(Y_t \mid X_t, \theta_t)^{\eta_\theta}.$$

To see the particle-filter approximation explicitly, suppose the predictive prior is represented by particles $\{(\theta^{(i)}_{t,-}, w^{(i)}_{t,-})\}_{i=1}^N$. We restrict the candidate posterior to the same particle support and optimize only over the updated weights $\omega = (\omega_1, \ldots, \omega_N) \in \Delta_N$. The finite-particle KL update is

$$\min_{\omega \in \Delta_N} \left\{ -\eta_\theta \sum_{i=1}^N \omega_i \log p_{\mathrm{proj}}(Y_t \mid X_t, \theta^{(i)}_{t,-}) + \sum_{i=1}^N \omega_i \log \frac{\omega_i}{w^{(i)}_{t,-}} \right\}.$$

The first-order condition over the simplex gives

$$\omega_i \propto w^{(i)}_{t,-} p_{\mathrm{proj}}(Y_t \mid X_t, \theta^{(i)}_{t,-})^{\eta_\theta}, \qquad \sum_{i=1}^N \omega_i = 1,$$

which is exactly the projected particle-filter weight update (8). Propagation through $p(\theta_t \mid \theta_{t-1})$ produces the predictive particle cloud, and resampling is the standard SMC step used to control weight degeneracy. □

When $\Sigma_{\theta,t} = \sigma_t^2 I$, the projected likelihood satisfies

$$-\log p_{\mathrm{proj}}(Y_t \mid X_t, \theta) = \frac{1}{2\sigma_t^2} \sum_{k=1}^{K_t} \{y_{t,k} - y_s(x_{t,k}, \theta)\}^2 + c_t,$$

where c_t does not depend on θ. Therefore minimizing the negative projected log likelihood is equivalent to minimizing the empirical projected squared-loss criterion on the current batch. If the inputs are sampled from the local design distribution F_X, this empirical loss is an unbiased batch analogue of the projected calibration objective up to observation-noise variance.

B.3 Recursive Gaussian discrepancy update

Before proving Proposition 2, we spell out the objects in the simplest expanding-support case. Suppose the previous discrepancy posterior is carried on support S_{t-1} and the current batch inputs enlarge the support to $S_t = S_{t-1} \cup X_t$. Then

$$u_t = \delta(S_t)$$

is the vector of discrepancy values on the enlarged support. The pre-update Gaussian prior

$$\bar{q}^\delta_{t|t-1}(u_t) = \mathcal{N}(a_t, P_t)$$

is obtained by GP conditioning from the previous posterior. The matrix G_t maps the carried state u_t to discrepancy values at the current inputs. In the expanding-support case, G_t is simply the selection matrix that extracts the coordinates of u_t corresponding to X_t. The residual covariance R_t is the covariance in the residual likelihood; it is not the GP kernel. It contains observation noise, nugget regularization, and any approximation variance included in the residual model.

Proof of Proposition 2. Let the predictive prior be

$$\bar{q}^{\delta}_{t|t-1}(u_t) = \mathcal{N}(a_t, P_t)$$

and the residual likelihood be

$$r_t \mid u_t \sim \mathcal{N}(G_t u_t, R_t).$$

The KL-regularized update has the tilted Bayesian form

$$q^{\delta}_t(u_t) \propto \bar{q}^{\delta}_{t|t-1}(u_t) p(r_t \mid u_t)^{\eta_\delta}.$$

Using the Gaussian densities,

$$\bar{q}^{\delta}_{t|t-1}(u_t) \propto \exp\left\{-\frac{1}{2}(u_t - a_t)^\top P_t^{-1}(u_t - a_t)\right\},$$

and

$$p(r_t \mid u_t)^{\eta_\delta} \propto \exp\left\{-\frac{\eta_\delta}{2}(r_t - G_t u_t)^\top R_t^{-1}(r_t - G_t u_t)\right\}.$$

Therefore,

$$q^{\delta}_t(u_t) \propto \exp\left\{-\frac{1}{2}\|u_t - a_t\|^2_{P_t^{-1}} - \eta_\delta \frac{1}{2}\|r_t - G_t u_t\|^2_{R_t^{-1}}\right\}.$$

Equivalently, the posterior mean minimizes the quadratic objective

$$\frac{1}{2}\|u_t - a_t\|^2_{P_t^{-1}} + \eta_\delta \frac{1}{2}\|r_t - G_t u_t\|^2_{R_t^{-1}}.$$

Expanding the quadratic terms in u_t gives posterior precision

$$C_t^{-1} = P_t^{-1} + \eta_\delta G_t^\top R_t^{-1} G_t$$

and posterior information vector

$$C_t^{-1} m_t = P_t^{-1} a_t + \eta_\delta G_t^\top R_t^{-1} r_t.$$

Thus

$$m_t = C_t\left(P_t^{-1} a_t + \eta_\delta G_t^\top R_t^{-1} r_t\right),$$

which gives the mean and covariance in Proposition 2. The same quadratic objective also gives the proximal characterization stated there. □

B.4 Discrepancy state maps for BRPC-E, BRPC-P, and BRPC-F

This section spells out the objects u_t, G_t, and $\bar{q}^{\delta}_{t|t-1}$ used in the generic discrepancy update.

BRPC-E: expanding-support exact recursion. Let S_{t-1} be the previous support and suppose

$$q^{\delta}_{t-1}(\delta_{S_{t-1}}) = \mathcal{N}(m_{t-1}, C_{t-1}).$$

After receiving inputs X_t, define the expanded support

$$S_t = S_{t-1} \cup X_t.$$

Under the GP prior, the conditional distribution of $\delta(S_t)$ given $\delta(S_{t-1})$ is Gaussian with conditional mean map

$$A_t = K_{S_t S_{t-1}} K^{-1}_{S_{t-1} S_{t-1}}$$

and conditional covariance

$$Q^E_t = K_{S_t S_t} - K_{S_t S_{t-1}} K^{-1}_{S_{t-1} S_{t-1}} K_{S_{t-1} S_t}.$$

Marginalizing over the previous posterior gives the transported predictive prior

$$\bar{q}^{\delta}_{t|t-1}(\delta_{S_t}) = \mathcal{N}(a_t, P_t), \qquad a_t = A_t m_{t-1}, \qquad P_t = Q^E_t + A_t C_{t-1} A_t^\top.$$

In BRPC-E, the discrepancy state is

$$u_t = \delta(S_t).$$

The matrix $G_t = H_t$ is the coordinate-selection matrix that extracts the entries of $\delta(S_t)$ corresponding to the current inputs X_t. Hence

$$r_t \mid \delta(S_t) \sim \mathcal{N}(H_t \delta(S_t), R_t),$$

and Proposition 2 gives the BRPC-E posterior.

BRPC-F: fixed-support recursion. Let $Z = \{z_1, \ldots, z_M\}$ be a fixed support and define

$$u_t = \delta(Z).$$

For current inputs X_t, the GP conditional-mean map from the fixed support to the current batch is

$$G_t = K_{X_t Z} K_{ZZ}^{-1}.$$

This is the "GP interpolation matrix" used in the main text: it maps discrepancy values on Z to the conditional mean of discrepancy values at X_t,

$$\mathbb{E}\{\delta(X_t) \mid \delta(Z) = u_t\} = G_t u_t.$$

The conditional GP variance at X_t given Z is

$$Q^F_t = K_{X_t X_t} - K_{X_t Z} K_{ZZ}^{-1} K_{Z X_t}.$$

If this uncertainty is kept explicit, the residual likelihood can be written as

$$r_t \mid u_t \sim \mathcal{N}(G_t u_t,\ R_t + Q^F_t).$$

Equivalently, one may absorb Q^F_t into an effective residual covariance and write the likelihood in the generic form

$$r_t \mid u_t \sim \mathcal{N}(G_t u_t, \widetilde{R}_t).$$

Thus R_t in the main text is not the GP kernel. It is the covariance used in the residual likelihood: it may include observation noise, nugget terms, and any approximation variance retained from the GP prediction step. The GP kernel determines G_t, Q^F_t, and the predictive prior over u_t.

In the exact expanding-support construction with independent Gaussian observation noise, $R_t = \sigma^2 I_{K_t}$. More generally, R_t denotes the residual-likelihood covariance. It may include observation noise, nugget regularization, anchor uncertainty from the shared residual approximation, or the GP conditional variance induced by a fixed-support approximation. The GP kernel determines the state transport and interpolation maps; R_t is the noise covariance used when assimilating the current residual batch.

BRPC-P: proxy-memory recursion. BRPC-P represents the carried discrepancy posterior through proxy observations. Suppose the carried posterior on support H is

$$q^{\delta}_H(\delta_H) = \mathcal{N}(m_H, C_H), \qquad p_0(\delta_H) = \mathcal{N}(0, K_H).$$

The goal is to replace the posterior q^{δ}_H by a synthetic Gaussian likelihood that, when combined with the original GP prior, gives the same Gaussian posterior. Introduce proxy observations

$$\tilde{y}_H \mid \delta_H \sim \mathcal{N}(\delta_H, \Lambda_H).$$

Under the prior $p_0(\delta_H) = \mathcal{N}(0, K_H)$ and this proxy likelihood, Gaussian conditioning gives posterior precision

$$K_H^{-1} + \Lambda_H^{-1}$$

and posterior information vector

$$\Lambda_H^{-1} \tilde{y}_H.$$

To reproduce the carried posterior $\mathcal{N}(m_H, C_H)$, whose precision and information vector are C_H^{-1} and $C_H^{-1} m_H$, respectively, it is therefore enough to choose

$$K_H^{-1} + \Lambda_H^{-1} = C_H^{-1}, \qquad \Lambda_H^{-1} \tilde{y}_H = C_H^{-1} m_H.$$

Equivalently,

$$\Lambda_H^{-1} = C_H^{-1} - K_H^{-1}, \qquad \Lambda_H^{-1} \tilde{y}_H = C_H^{-1} m_H.$$

Thus the proxy observations carry exactly the same Gaussian information as the posterior they replace: updating the GP prior with the proxy likelihood recovers $q_H^{\delta}(\delta_H) = \mathcal{N}(m_H, C_H)$.

This ideal construction requires $C_H^{-1} - K_H^{-1} \succeq 0$, so that Λ_H is a valid covariance matrix. In finite precision or when the carried posterior is only approximately represented, we use a stabilized proxy construction by adding nugget regularization, covariance inflation, or spectral clipping to keep Λ_H positive definite. These stabilized variants are not exactly algebraically equivalent to BRPC-E, but they preserve the same proxy-memory interpretation while improving numerical robustness.

Proposition 5 (Relationship among BRPC-E, BRPC-P, and BRPC-F)**.** *Under the same Gaussian prior, likelihood, and support, ideal BRPC-P is algebraically equivalent to BRPC-E. BRPC-F is the corresponding projected-state approximation: it satisfies the same KL-regularized update form in a fixed-support state class, and its approximation gap to a full discrepancy comparator is controlled by fixed-support projection error.*

The proof is given in Appendix E.4.

B.5 Particle-specific and shared residual constructions

The most direct Bayesian analogue of static BPC is particle-specific. If the projected parameter posterior is represented by particles $\{\theta_t^{(i)}, w_t^{(i)}\}_{i=1}^N$, then particle i induces residuals

$$r_t^{(i)} = Y_t - y_s(X_t, \theta_t^{(i)}).$$

A fully conditional implementation attaches a discrepancy posterior $q_t^{\delta,(i)}$ to each particle and applies Proposition 2 separately to each residual batch $r_t^{(i)}$.

In the main experiments, we use a shared expert-level residual construction. Let

$$\bar{\mu}_t(X_t) = \sum_{i=1}^{N} w_t^{(i)} y_s(X_t, \theta_t^{(i)})$$

be the particle-weighted simulator mean. The shared residual batch is

$$r_t = Y_t - \bar{\mu}_t(X_t).$$

The discrepancy posterior is then updated once per expert using this shared residual batch. This approximation reduces computation and avoids maintaining many adaptive discrepancy states. The state-sharing ablation in Appendix J.4 shows that the particle-specific construction provides no systematic empirical gain in our benchmark while increasing runtime and evidence-stream variability.

C Residual Re-Anchoring Construction

For expert e at time t, let

$$\bar{\mu}_{e,t}(x) := \mathbb{E}_{q_{e,t}^{\theta}}\big[y_s(x, \theta)\big] \tag{23}$$

denote the current simulator anchor. The re-anchored residual dataset is $\mathcal{D}_{e,t}^{\mathrm{res}} := \big\{\big(x_{u,k},\, y_{u,k} - \bar{\mu}_{e,t}(x_{u,k})\big)\big\}_{(u,k)\in\mathcal{I}_{e,t}}$, where $\mathcal{I}_{e,t}$ indexes the observations currently assigned to the expert segment. Under the Gaussian discrepancy model, BRPC-RRA fits the local discrepancy state by

$$q_{e,t}^{\mathrm{RRA}}(\delta) \propto p(\mathcal{D}_{e,t}^{\mathrm{res}} \mid \delta)\, p_0(\delta). \tag{24}$$

Computational and statistical role of RRA. BRPC-RRA should be interpreted as a segment-level re-anchoring variant rather than a purely recursive discrepancy update. Unlike BRPC-E/P/F, it refits the local discrepancy posterior for the active segment at each batch after recomputing archived residuals relative to the current simulator anchor. This is computationally heavier, especially when evaluating the simulator at archived inputs is expensive and cannot be cached.

The statistical benefit is that the residual dataset is made coherent with a single current calibration anchor. This reduces residual-anchor heterogeneity caused by carrying discrepancy information from older anchors, and can produce more segment-local predictive evidence for BOCPD. In this sense, BOCPD acts not only as a changepoint detector but also as a memory-selection mechanism: it selects which historical residuals remain in the active segment. This is similar in spirit to maintaining a replay buffer in online learning, except that the stored observations are re-expressed through the current simulator anchor before the discrepancy model is refit.

This distinction also explains the different behavior of RRA under jumps and drift. In piecewise-constant or jump-dominated settings, a correctly selected segment gives a nearly stationary residual dataset under the current anchor, so RRA can provide clean BOCPD-facing evidence. In drifting settings, however, the residual target changes continuously, and RRA relies more heavily on BOCPD to choose a useful segment length. Faster drift can therefore lead to shorter selected segments and more frequent restarts. This is why BRPC-RRA is best viewed as a jump-oriented state-side response, while recursive BRPC variants with score-based restart are more robust defaults under heterogeneous nonstationarity.

D Predictive-law details

Particle-specific predictive law. For the particle-specific formulation, conditional on particle $\theta_t^{(i)}$ and discrepancy posterior $q_t^{\delta,(i)}(\delta_{S_t^{(i)}}) = \mathcal{N}(m_t^{(i)}, C_t^{(i)})$, define

$$A_{\star,t}^{(i)} = K_{X_\star, S_t^{(i)}} K_{S_t^{(i)} S_t^{(i)}}^{-1}, \qquad Q_{\star,t}^{(i)} = K_{X_\star X_\star} - K_{X_\star, S_t^{(i)}} K_{S_t^{(i)} S_t^{(i)}}^{-1} K_{S_t^{(i)}, X_\star}.$$

Then

$$\begin{aligned} Y_\star \mid X_\star, \theta_t^{(i)}, q_t^{\delta,(i)} \sim \mathcal{N}\Big(&\mu_s(X_\star, \theta_t^{(i)}) + A_{\star,t}^{(i)} m_t^{(i)}, \\ &\Sigma_s(X_\star, \theta_t^{(i)}) + Q_{\star,t}^{(i)} + A_{\star,t}^{(i)} C_t^{(i)} (A_{\star,t}^{(i)})^\top + \sigma^2 I\Big). \end{aligned} \tag{25}$$

Hence the expert-level predictive law is the particle mixture

$$p_e(Y_\star \mid X_\star) \approx \sum_{i=1}^{N} w_{e,t}^{(i)} \mathcal{N}\left(Y_\star; \mu_{e,\star}^{(i)}, \Sigma_{e,\star}^{(i)}\right), \tag{26}$$

where $\mu_{e,\star}^{(i)}$ and $\Sigma_{e,\star}^{(i)}$ are the mean and covariance in (25).

Shared BRPC-P predictive law. In the ideal proxy construction, BRPC-P is algebraically equivalent to BRPC-E, so it induces the same shared expert-level predictive law as the exact support-to-batch transport construction. For stabilized proxy memories, the same form is used with the approximate posterior

$$q_t^{\delta,\mathrm{P}}(\delta_{S_t}) = \mathcal{N}(m_t^{\mathrm{P}}, C_t^{\mathrm{P}}).$$

BRPC-F predictive law. For BRPC-F, the discrepancy state is

$$u_t = \delta(Z), \qquad q_t^u(u_t) = \mathcal{N}(m_t^u, C_t^u),$$

on a fixed support $Z = \{z_1, \ldots, z_M\}$. Let

$$G_\star = K_{X_\star, Z} K_{ZZ}^{-1}, \qquad \widetilde{Q}_\star = K_{X_\star X_\star} - K_{X_\star, Z} K_{ZZ}^{-1} K_{Z, X_\star}. \tag{27}$$

Then

$$\delta(X_\star) \mid q_t^u \sim \mathcal{N}\left(G_\star m_t^u,\ \widetilde{Q}_\star + G_\star C_t^u G_\star^\top\right), \tag{28}$$

and

$$p_e(Y_\star \mid X_\star) \approx \sum_{i=1}^{N} w_{e,t}^{(i)} \mathcal{N}\left(Y_\star; \mu_s(X_\star, \theta_{e,t}^{(i)}) + G_\star m_t^u,\ \Sigma_s(X_\star, \theta_{e,t}^{(i)}) + \widetilde{Q}_\star + G_\star C_t^u G_\star^\top + \sigma^2 I\right). \tag{29}$$

E Proofs for BRPC

E.1 From the KL update to the finite-dimensional BRPC-E objective

Let the predictive prior over the carried discrepancy state be

$$q_{t|t-1}(z) = \mathcal{N}(a_t, P_t), \qquad M_t = P_t^{-1}.$$

In this subsection we specialize the generic discrepancy update to BRPC-E. The state is $z = \delta(S_t)$ on the expanded support $S_t = S_{t-1} \cup X_t$, and H_t is the coordinate-selection matrix that extracts the entries of $\delta(S_t)$ corresponding to the current batch inputs X_t. Thus, in the notation of Proposition 2, $G_t = H_t$ for BRPC-E. Consider the KL-regularized variational update

$$q_t = \arg\min_q \left\{ \eta\, \mathbb{E}_q \left[\frac{1}{2} \|r_t - H_t z\|^2_{R_t^{-1}} \right] + \mathrm{KL}(q \,\|\, q_{t|t-1}) \right\}. \tag{30}$$

Restricting to Gaussian posteriors $q = \mathcal{N}(m, C)$, the objective equals

$$\begin{aligned} &\eta \frac{1}{2} \|r_t - H_t m\|^2_{R_t^{-1}} + \eta \frac{1}{2} \mathrm{Tr}(H_t^\top R_t^{-1} H_t C) \\ &\quad + \frac{1}{2} \left[\mathrm{Tr}(P_t^{-1} C) + (m - a_t)^\top P_t^{-1} (m - a_t) - d_t + \log \frac{\det P_t}{\det C} \right]. \end{aligned} \tag{31}$$

Optimizing over C and m yields

$$C_t^{-1} = P_t^{-1} + \eta H_t^\top R_t^{-1} H_t,$$

and

$$m_t = C_t \left(P_t^{-1} a_t + \eta H_t^\top R_t^{-1} r_t \right).$$

The mean m_t is equivalently the minimizer of

$$z_t = \arg\min_z \left\{ \eta \frac{1}{2} \|r_t - H_t z\|^2_{R_t^{-1}} + \frac{1}{2} \|z - a_t\|^2_{M_t} \right\}.$$

This proves the equivalence between the KL-regularized Gaussian update and the BRPC-E objective.

E.2 Proof of Theorem 1

Fixed-support example. For intuition, consider BRPC-F with a fixed support $Z = \{z_1, \ldots, z_M\}$. The discrepancy state is $u_t = \delta(Z) \in \mathbb{R}^M$. Since the support does not change, the transport map is $A_t = I$, so the pre-update mean is $a_t = z_{t-1}$, possibly with inflated covariance $P_t = C_{t-1} + Q_t$. For a new batch X_t, the GP interpolation map is $G_t = K_{X_t Z} K_{ZZ}^{-1}$, and the residual likelihood is $r_t \mid u_t \sim \mathcal{N}(G_t u_t, R_t)$, with $R_t = \sigma^2 I$ in the simplest case. Thus the update

$$z_t = \arg\min_z \left\{ \eta_\delta \frac{1}{2} \|r_t - G_t z\|^2_{R_t^{-1}} + \frac{1}{2} \|z - z_{t-1}\|^2_{M_t} \right\}$$

is a proximal residual-fitting step on the fixed discrepancy state. The theorem compares this algorithmic state z_t with an arbitrary reference state $v_t = \delta_t^\star(Z)$. In the fixed-support case, the path-variation term is $\sum_t \|v_t - v_{t-1}\|^2_{M_t}$; for expanding support, A_t replaces the identity and transports the comparator between state spaces.

Proof. The first-order optimality condition for (16) is

$$\eta \nabla \ell_t(z_t) + M_t(z_t - a_t) = 0. \tag{32}$$

Since ℓ_t is convex,

$$\ell_t(z_t) - \ell_t(u_t) \le \langle \nabla \ell_t(z_t), z_t - u_t \rangle. \tag{33}$$

Multiplying by η and using (32),

$$\eta \left[\ell_t(z_t) - \ell_t(u_t) \right] \le \langle M_t(z_t - a_t), u_t - z_t \rangle. \tag{34}$$

Using the weighted three-point identity

$$2\langle a - b, c - a \rangle_M = \|c - b\|^2_M - \|c - a\|^2_M - \|a - b\|^2_M,$$

with $a = z_t$, $b = a_t$, $c = u_t$, and $M = M_t$, gives

$$\eta\,[\ell_t(z_t) - \ell_t(u_t)] \le \frac{1}{2}\|u_t - a_t\|_{M_t}^2 - \frac{1}{2}\|u_t - z_t\|_{M_t}^2 - \frac{1}{2}\|z_t - a_t\|_{M_t}^2. \tag{35}$$

Since $a_t = A_t z_{t-1}$,

$$u_t - a_t = (u_t - A_t u_{t-1}) + A_t(u_{t-1} - z_{t-1}).$$

For any $\beta > 0$,

$$\begin{aligned}\|u_t - a_t\|_{M_t}^2 &\le (1+\beta)\|u_t - A_t u_{t-1}\|_{M_t}^2 + \left(1 + \frac{1}{\beta}\right)\|A_t(u_{t-1} - z_{t-1})\|_{M_t}^2 \\ &\le (1+\beta)\|u_t - A_t u_{t-1}\|_{M_t}^2 + \left(1 + \frac{1}{\beta}\right)\gamma\|u_{t-1} - z_{t-1}\|_{M_{t-1}}^2, \end{aligned} \tag{36}$$

where the second line uses Assumption 1. Choose

$$\beta = \frac{\gamma}{1-\gamma}.$$

Then

$$1 + \beta = \frac{1}{1-\gamma}, \qquad \left(1 + \frac{1}{\beta}\right)\gamma = 1.$$

Substituting into (35) yields

$$\eta\,[\ell_t(z_t) - \ell_t(u_t)] \le \frac{1}{2(1-\gamma)}\|u_t - A_t u_{t-1}\|_{M_t}^2 + \frac{1}{2}\|u_{t-1} - z_{t-1}\|_{M_{t-1}}^2 - \frac{1}{2}\|u_t - z_t\|_{M_t}^2. \tag{37}$$

Summing over $t = 1, \dots, T$ telescopes:

$$\eta \sum_{t=1}^{T} [\ell_t(z_t) - \ell_t(u_t)] \le \frac{1}{2}\|u_0 - z_0\|_{M_0}^2 + \frac{1}{2(1-\gamma)} \sum_{t=1}^{T} \|u_t - A_t u_{t-1}\|_{M_t}^2.$$

Dividing by η proves (18). □

E.3 A sufficient condition for transport contraction

Proposition 6 (Sufficient condition for Assumption 1)**.** *Suppose*

$$P_t = A_t C_{t-1} A_t^\top + Q_t, \qquad Q_t \succeq q_{\min} I,$$

and assume

$$A_t^\top A_t \preceq a_{\max} I, \qquad C_{t-1} \preceq c_{\max} I.$$

If

$$\gamma = \frac{a_{\max} c_{\max}}{q_{\min}} < 1,$$

then Assumption 1 holds.

Proof. Since $P_t \succeq Q_t \succeq q_{\min} I$,

$$M_t = P_t^{-1} \preceq q_{\min}^{-1} I.$$

Therefore

$$A_t^\top M_t A_t \preceq q_{\min}^{-1} A_t^\top A_t \preceq \frac{a_{\max}}{q_{\min}} I.$$

Since $C_{t-1} \preceq c_{\max} I$,

$$M_{t-1} = C_{t-1}^{-1} \succeq c_{\max}^{-1} I.$$

Thus

$$\frac{a_{\max}}{q_{\min}} I \preceq \frac{a_{\max} c_{\max}}{q_{\min}} M_{t-1} = \gamma M_{t-1}.$$

Combining the two inequalities gives

$$A_t^\top M_t A_t \preceq \gamma M_{t-1}.$$

□

Interpretation. Proposition 6 shows that Assumption 1 can be encouraged by adding a small amount of uncertainty when transporting discrepancy information forward. For example, one may use a predictive covariance of the form

$$P_t = A_t C_{t-1} A_t^\top + Q_t^\delta, \qquad Q_t^\delta \succeq q_{\min} I,$$

or, in the fixed-support case,

$$P_t = C_{t-1} + \sigma_{\delta,\mathrm{rw}}^2 I.$$

This process-noise or covariance-inflation term prevents the carried discrepancy posterior from becoming overconfident and treating historical residual information as an infinitely precise constraint on future batches. If the transported precision is too large, the contraction condition may fail and the recursive discrepancy state can become overly rigid or produce overly sharp pre-update predictive evidence. Thus the condition is best understood as an algorithmic memory-stability requirement, not as a stochastic assumption on the data-generating process.

E.4 Proof of Proposition 5

Proof. We first compare BRPC-E and ideal BRPC-P. Fix the same support H and the same Gaussian prior

$$p_0(\delta_H) = \mathcal{N}(0, K_H).$$

Suppose the carried posterior on H is

$$q_H^\delta(\delta_H) = \mathcal{N}(m_H, C_H).$$

BRPC-P constructs proxy observations

$$\tilde{y}_H \mid \delta_H \sim \mathcal{N}(\delta_H, \Lambda_H)$$

so that

$$\Lambda_H^{-1} = C_H^{-1} - K_H^{-1}, \qquad \Lambda_H^{-1}\tilde{y}_H = C_H^{-1} m_H.$$

Under the GP prior and this proxy likelihood, the posterior precision is

$$K_H^{-1} + \Lambda_H^{-1} = C_H^{-1},$$

and the posterior information vector is

$$\Lambda_H^{-1}\tilde{y}_H = C_H^{-1} m_H.$$

Therefore the proxy likelihood reproduces exactly the same Gaussian posterior $\mathcal{N}(m_H, C_H)$ as the carried BRPC-E posterior. Since subsequent assimilation of the current residual batch is Gaussian and depends only on the prior precision and information vector, BRPC-E and ideal BRPC-P produce the same posterior mean and covariance under exact arithmetic and positive-definiteness of Λ_H.

We next consider BRPC-F. In BRPC-F, the discrepancy process is represented by its values on a fixed support Z:

$$u_t = \delta(Z).$$

For current inputs X_t, let

$$G_t = K_{X_t Z} K_{ZZ}^{-1}$$

be the GP conditional-mean map from the fixed support to the batch inputs. The residual model becomes

$$r_t \mid u_t \sim \mathcal{N}(G_t u_t, R_t),$$

possibly with fixed-support conditional variance absorbed into R_t. Therefore the BRPC-F update is

$$u_t = \arg\min_u \left\{ \eta_\delta \frac{1}{2} \|r_t - G_t u\|_{R_t^{-1}}^2 + \frac{1}{2} \|u - A_t u_{t-1}\|_{M_t}^2 \right\},$$

which is the same KL-regularized recursive update, but restricted to the fixed-support state class.

Finally, let f_t be any full discrepancy comparator and define its best fixed-support projection on the current batch by

$$u_t^\sharp \in \arg\min_u \|f_t(X_t) - G_t u\|_{R_t^{-1}}^2.$$

Then BRPC-F competes with $u_t^\sharp$ within the fixed-support class, and the additional loss relative to the full comparator is governed by the projection error

$$\|f_t(X_t) - G_t u_t^\sharp\|_{R_t^{-1}}^2.$$

Thus BRPC-F is a projected-state approximation to the full recursive discrepancy update, with approximation quality determined by the fixed-support projection error. □

F Controller–Learner Compatibility Theory

This appendix collects auxiliary theory used to interpret restart and controller–learner compatibility. Appendix F.1 formalizes the operational distinction among drift, regime change, and mixed nonstationarity. Appendix F.2 gives a segmented surrogate-regret view of BOCPD-controlled BRPC: the BOCPD mixture can compete with segmented expert paths when the experts provide meaningful prequential evidence. Appendix F.3 gives a posterior-odds testing view of hard restart, which is the perspective used in the main text to explain controller–learner compatibility. Appendix F.4 records the finite-dimensional identifiability calculation behind the joint-assimilation baseline. Appendix F.5 interprets restart as adaptive residual-data selection for re-anchored discrepancy learning, and Appendix F.6 collects the auxiliary wCUSUM and re-anchoring results.

F.1 Drift, Regime Change, and Mixed Nonstationarity

This section formalizes the operational distinction between drift and regime change used in the experiments and controller discussion. The distinction is not defined by whether a reset is triggered. Rather, it is defined by how the local calibration problem evolves over time.

Let g_t denote the local calibration object at batch t. In the simplest case, g_t is the projected calibration target $\theta_t^\dagger$; more generally, it may include both the projected parameter target and the effective residual target seen by the discrepancy learner. Let A_t denote the transport map used to carry the previous local state forward, and let $\|\cdot\|_{G,t}$ be a problem-dependent local metric, such as an M_t-weighted state norm or an $L^2(F_X)$ response metric.

Within-regime drift. An interval $I = \{\tau, \dots, T\}$ is a drift regime if the transported one-step variation is small or accumulates gradually:

$$\Delta_t^{\text{drift}} := \|g_t - A_t g_{t-1}\|_{G,t}, \qquad \sum_{t \in I} (\Delta_t^{\text{drift}})^2 \text{ is moderate.}$$

Thus the local calibration problem may move over time, but consecutive batches remain close enough that carrying forward recent calibration and discrepancy information is beneficial.

Regime change. A time τ is a regime change if the local calibration problem changes abruptly or structurally relative to the preceding regime. This can occur through a jump in the projected calibration target, a change in the residual process, a shift in operating conditions, or a change in the simulator–reality relationship. Formally, such a change corresponds to a transported variation

$$\Delta_\tau^{\text{jump}} := \|g_\tau - A_\tau g_{\tau-1}\|_{G,\tau}$$

that is large relative to the within-regime drift scale or the predictive uncertainty of the continuing local learner. After such a change, information from the previous regime can introduce bias into the new local calibration problem.

Mixed nonstationarity. A stream is mixed when it contains both gradual within-regime drift and a finite number of abrupt regime changes:

$$\{1, \dots, T\} = \bigcup_{s=1}^{S} I_s,$$

where each segment I_s satisfies the drift condition and adjacent segments are separated by regime changes.

This distinction explains the event-level restart metrics used in the experiments. Precision, recall, and delay are evaluated against annotated abrupt regime changes, because these metrics measure whether the restart mechanism responds near structural changes. In drifting settings there are no annotated abrupt changes. In mixed settings, a restart caused by gradual drift may still help local prediction, but it is not counted as a true positive for regime-change detection.

F.2 Segmented Surrogate Regret for B-BRPC

This section connects the shared Exact discrepancy recursion to the pre-update expert evidence consumed by Restart-BOCPD. The result is intentionally a surrogate regret statement: it controls the BOCPD mixture loss relative to segmented mean-level discrepancy comparators, while keeping covariance and normalization terms as path-dependent offsets.

For an active Exact expert, let $x_t := a_t = A_t z_{t-1}$ denote the pre-update discrepancy state and let z_t denote the post-update state. Recall

$$\ell_t(z) = \frac{1}{2}\|r_t - H_t z\|^2_{R_t^{-1}}.$$

Assumption 2 (Pre/post regularity)**.** *For each t, there exist constants $\rho_t \geq 0$ and $G_t \geq 0$ such that*

$$\eta H_t^\top R_t^{-1} H_t \preceq \rho_t M_t, \qquad \|\nabla \ell_t(v)\|_{M_t^{-1}} \leq G_t$$

for all relevant states v.

Proposition 7 (Pre/post update gap)**.** *Under Assumption 2,*

$$\ell_t(x_t) - \ell_t(z_t) \leq \Gamma_t, \qquad \Gamma_t := \eta\left(1 + \frac{\rho_t}{2}\right) G_t^2.$$

Corollary 1 (Pre-update quadratic tracking)**.** *Under Assumption 1 and Assumption 2,*

$$\sum_{t=1}^{T}\{\ell_t(x_t) - \ell_t(u_t)\} \leq \frac{1}{2\eta}\|u_0 - z_0\|^2_{M_0} + \frac{1}{2\eta(1-\gamma)}\mathrm{V}_T(u_{1:T}) + \sum_{t=1}^{T}\Gamma_t.$$

For a shared Exact expert e, suppose the pre-update predictive covariance decomposes as

$$\Sigma^{\mathrm{pre}}_{e,t} = \Sigma^{\mathrm{sim}}_{e,t} + H_{e,t}P_{e,t}H_{e,t}^\top + R_{e,t}, \qquad \Sigma^{\mathrm{sim}}_{e,t} \succeq 0, \quad P_{e,t} \succeq 0, \quad R_{e,t} \succ 0.$$

Let

$$\mu^{\mathrm{pre}}_{e,t} = \bar{\mu}_{e,t}(X_t) + H_{e,t}x_{e,t}$$

and define the prequential loss

$$L^{\mathrm{pre}}_{e,t} := -\log p^{\mathrm{pre}}_e(Y_t \mid X_t).$$

Then

$$L^{\mathrm{pre}}_{e,t} = \frac{1}{2}\|Y_t - \mu^{\mathrm{pre}}_{e,t}\|^2_{(\Sigma^{\mathrm{pre}}_{e,t})^{-1}} + \frac{1}{2}\log\det\Sigma^{\mathrm{pre}}_{e,t} + \frac{B_t}{2}\log(2\pi).$$

Lemma 1 (Covariance domination)**.** *Under the covariance decomposition above,*

$$\Sigma^{\mathrm{pre}}_{e,t} \succeq R_{e,t}, \qquad (\Sigma^{\mathrm{pre}}_{e,t})^{-1} \preceq R^{-1}_{e,t}.$$

Proposition 8 (Prequential loss upper bound)**.** *Let*

$$r_{e,t} := Y_t - \bar{\mu}_{e,t}(X_t), \qquad \ell^{\mathrm{prequad}}_{e,t}(x_{e,t}) := \frac{1}{2}\|r_{e,t} - H_{e,t}x_{e,t}\|^2_{R^{-1}_{e,t}}.$$

Then

$$L^{\mathrm{pre}}_{e,t} \leq c_{e,t} + \ell^{\mathrm{prequad}}_{e,t}(x_{e,t}), \qquad c_{e,t} := \frac{1}{2}\log\det\Sigma^{\mathrm{pre}}_{e,t} + \frac{B_t}{2}\log(2\pi).$$

Let Π_T be the set of admissible expert paths $\pi = (e_1, \ldots, e_T)$ and let $\Pi(\pi)$ denote the BOCPD prior weight induced by the restart hazard. Define the cumulative BOCPD mixture loss

$$L^{\mathrm{mix}}_{1:T} := -\log\left(\sum_{\pi\in\Pi_T}\Pi(\pi)\exp\left\{-\sum_{t=1}^{T}L^{\mathrm{pre}}_{e_t,t}\right\}\right).$$

Theorem 2 (Pathwise BOCPD mixture bound)**.** *For every expert path $\pi = (e_1, \ldots, e_T) \in \Pi_T$,*

$$L^{\mathrm{mix}}_{1:T} \leq \sum_{t=1}^{T}L^{\mathrm{pre}}_{e_t,t} - \log\Pi(\pi).$$

Now let the horizon be partitioned into segments $I_s = \{\tau_{s-1}+1, \dots, \tau_s\}$, with $0 = \tau_0 < \tau_1 < \cdots < \tau_S = T$. Let $\pi^\star = (e_1^\star, \dots, e_T^\star)$ be the expert path that follows these segments. For each segment s, let $\{u_t^{(s)}\}_{t \in I_s}$ be a comparator sequence in the shared Exact state space of the active expert on that segment, and define

$$\widetilde{L}_{e,t}(u) := c_{e,t} + \ell_{e,t}^{\text{prequad}}(u).$$

Theorem 3 (Combined Exact + BOCPD surrogate regret)**.** *Assume that, on every segment, the active Exact expert satisfies Assumption 1 and Assumption 2, and that its prequential predictive law satisfies the covariance decomposition above. Then*

$$\begin{aligned} & L_{1:T}^{\text{mix}} - \sum_{s=1}^{S} \sum_{t \in I_s} \widetilde{L}_{e_t^\star, t}\left(u_t^{(s)}\right) \\ & \leq -\log \Pi(\pi^\star) + \sum_{s=1}^{S} \frac{1}{2\eta} \left\| u_{\tau_{s-1}}^{(s)} - z_{\text{init}} \right\|_{M_{\tau_{s-1}}}^2 \\ & \quad + \sum_{s=1}^{S} \frac{1}{2\eta(1-\gamma)} \sum_{t \in I_s} \left\| u_t^{(s)} - A_t u_{t-1}^{(s)} \right\|_{M_t}^2 + \sum_{t=1}^{T} \Gamma_t. \end{aligned} \tag{38}$$

Here z_{init} is the expert's restart initialization state.

Interpretation. Theorem 3 should not be read as a full predictive-regret theorem against a comparator that also optimizes covariance. It is a segmented surrogate regret statement: the Exact inner recursion controls the pre-update quadratic mean-level loss, while the BOCPD mixture pays only the path complexity $-\log \Pi(\pi^\star)$ for following a segmented expert path.

Oracle interpretation. Theorem 3 implies an oracle inequality over restart schedules. To make this explicit, consider a constant restart hazard $h \in (0,1)$ and a candidate changepoint path π with $K(\pi)$ segment transitions over T batches. Then

$$-\log \Pi(\pi) = K(\pi) \log \frac{1}{h} + (T - K(\pi)) \log \frac{1}{1-h}.$$

Consequently,

$$L_{1:T}^{\text{mix}} \leq \min_{\pi \in \Pi_T} \left\{ \sum_{t=1}^{T} L_{e_t,t}^{\text{pre}} + K(\pi) \log \frac{1}{h} + (T - K(\pi)) \log \frac{1}{1-h} \right\}.$$

Combining this with Theorem 3 yields a segmented surrogate regret bound against candidate changepoint paths $\pi \in \Pi_T$ with $K(\pi)$ segment transitions over T batches. Thus Restart-BOCPD satisfies an oracle inequality in the standard prequential mixture sense, conditional on the candidate experts providing meaningful predictive evidence. This is a soft-mixture guarantee; it does not imply that the realized hard-reset MAP path is hindsight optimal.

F.3 BOCPD-Style Restart as Posterior-Odds Testing

This section gives a sequential-testing interpretation of the Restart-BOCPD rule. It complements the single-step restart-odds calculation in the main text. The results below describe when a BOCPD-style restart statistic is stable under idealized likelihood-ratio drift conditions, and why adaptive discrepancy learning can disrupt those conditions.

Let r_t denote the BOCPD run length and let $\pi_t(r) = \mathbb{P}(r_t = r \mid B_{1:t})$. Let r_{anc} be the run length of the anchor expert associated with the most recent hard restart, and define

$$p_{\text{anc}} := \pi_t(r_{\text{anc}}), \qquad p^\star := \max_{s > r_{\text{anc}}} \pi_t(t-s).$$

A hard restart is declared when

$$p^\star > (1+\eta)\, p_{\text{anc}}.$$

Lemma 2 (Posterior odds as a windowed log Bayes factor)**.** *Assume a constant hazard and factorized batch predictive likelihoods. For any candidate segment start s and anchor run length r_{anc},*

$$\log \frac{\pi_t(t-s)}{\pi_t(r_{\mathrm{anc}})} = \log \frac{\mathrm{PriorOdds}(s)}{\mathrm{PriorOdds}(r_{\mathrm{anc}})} + \sum_{u=s}^{t} \log \frac{q_s(B_u \mid \mathcal{F}_{u-1})}{q_{\mathrm{anc}}(B_u \mid \mathcal{F}_{u-1})}.$$

Thus the hard restart rule is equivalent to a threshold test on a windowed log-likelihood ratio, up to an additive prior-odds term.

Define a one-step log-likelihood-ratio increment

$$\Delta\ell_t := \log \frac{q_{\mathrm{new}}(B_t \mid \mathcal{F}_{t-1})}{q_{\mathrm{old}}(B_t \mid \mathcal{F}_{t-1})}, \qquad S_t := \sum_{u=1}^{t} \Delta\ell_u,$$

and let $\tau_h := \inf\{t \geq 1 : S_t \geq h\}$. Write

$$\mu_t := \mathbb{E}[\Delta\ell_t \mid \mathcal{F}_{t-1}], \qquad Z_t := \Delta\ell_t - \mu_t.$$

Assumption 3 (Conditional sub-Gaussian fluctuations)**.** *There exists $\nu_\ell^2 < \infty$ such that, for all $\lambda \in \mathbb{R}$,*

$$\mathbb{E}[\exp(\lambda Z_t) \mid \mathcal{F}_{t-1}] \leq \exp\left(\frac{\lambda^2 \nu_\ell^2}{2}\right).$$

Proposition 9 (Finite-horizon restart testing bounds)**.** *Under Assumption 3:*

(a) If $\mu_t \leq \bar{\mu}_0$ for all $t \leq T$, then for any $h > T\bar{\mu}_0$,

$$\mathbb{P}\left(\sup_{1 \leq t \leq T} S_t \geq h\right) \leq \exp\left(-\frac{(h - T\bar{\mu}_0)^2}{2\nu_\ell^2 T}\right).$$

(b) If a change occurs at time ν and there exists $\mu_1 > 0$ such that $\mu_t \geq \mu_1$ for all $t > \nu$, then the detection delay obeys the standard positive-drift scaling

$$\mathbb{E}[(\tau_h - \nu)_+] \lesssim \frac{h}{\mu_1} + \frac{\nu_\ell^2}{\mu_1^2},$$

up to universal constants from the chosen sub-Gaussian hitting-time inequality.

Proof sketch. Part (a) applies a maximal sub-Gaussian concentration inequality to the centered process $\sum_{u=1}^{t} Z_u$ after subtracting the deterministic drift upper bound $T\bar{\mu}_0$. Part (b) follows from the corresponding positive-drift hitting-time argument: once the candidate accumulates systematic predictive advantage at rate at least μ_1, the threshold h is crossed after $O(h/\mu_1)$ steps, with an additive fluctuation penalty of order ν_ℓ^2/μ_1^2. □

Interpretation. Restart is stable within a regime when candidate experts do not accumulate systematic predictive advantage over the anchor, i.e. when the drift of the candidate-versus-anchor log-likelihood ratio is nonpositive or small. After a jump, restart is fast only if a fresh candidate obtains positive predictive drift before the old expert adapts away the mismatch. This is where controller–learner compatibility enters: a strongly adaptive local learner can change the drift conditions seen by BOCPD, either by creating excessive within-regime surprise or by absorbing post-jump mismatch.

F.4 Identifiability Interpretation of Joint Assimilation

This section records a simple finite-dimensional calculation explaining why a joint assimilation channel can blur attribution between calibration parameters and model discrepancy.

Consider the Gaussian regression model

$$y = y_s(x, \theta) + \Phi_m(x)^\top \beta + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2),$$

where $\Phi_m(x) = (\phi_1(x), \ldots, \phi_m(x))^\top$ spans a finite-dimensional discrepancy class. Let $\psi = (\theta, \beta)$ and partition the Fisher information as

$$I(\psi) = \begin{pmatrix} I_{\theta\theta} & I_{\theta\beta} \\ I_{\beta\theta} & I_{\beta\beta} \end{pmatrix}.$$

Lemma 3 (Structural confounding)**.** *If $I_{\beta\beta} \succ 0$, then the profiled Fisher information for θ after eliminating β is*

$$I_m(\theta) = I_{\theta\theta} - I_{\theta\beta} I_{\beta\beta}^{-1} I_{\beta\theta} \preceq I_{\theta\theta}.$$

Moreover, there is information loss in at least one direction whenever $I_{\theta\beta} \neq 0$, i.e. whenever some simulator sensitivity direction has nonzero projection onto the discrepancy span under the design distribution.

Proof. Under the Gaussian model, the Fisher information is the Gram matrix of the score functions for (θ, β). Profiling out β gives the Schur complement. Since $I_{\theta\beta} I_{\beta\beta}^{-1} I_{\beta\theta} \succeq 0$, the profiled information is no larger than $I_{\theta\theta}$ in Loewner order. If $I_{\theta\beta} \neq 0$, this positive-semidefinite correction is nonzero, so at least one direction of θ loses information. □

Connection to BOCPD-WardPFMove. The BOCPD-controlled filtering baseline updates parameters through a joint assimilation channel. Under simulator misspecification, the same residual can be explained by parameter movement, discrepancy-like bias, observation noise, or regime change. Lemma 3 formalizes the static version of this attribution problem: flexible discrepancy directions can reduce the information available for identifying θ. In the online setting, this appears as inaccurate tracking together with weak hard-restart evidence, because the local filter can smooth part of a regime change into parameter motion before BOCPD sees a clean likelihood-ratio contrast.

F.5 Restart as Adaptive Residual-Data Selection

This section gives a simple interpretation of why segment-local residual refitting can help discrepancy learning. Even when the ground-truth discrepancy is fixed, the residual target seen by the discrepancy learner is generally nonstationary because residuals depend on the calibration anchor and on the selected training segment.

Let τ be the most recent restart time. For an anchor sequence $\bar{\theta}_{\tau:t-1}$, define the effective residual target at historical batch u by

$$r_u^\star(x; \bar{\theta}_u) := y_s(x, \theta_u^\star) + \delta^\star(x) - y_s(x, \bar{\theta}_u).$$

Define the residual-target heterogeneity over the current segment by

$$\Gamma_r(\tau, t-1) := \sum_{u=\tau}^{t-2} \left\| r_{u+1}^\star(\cdot; \bar{\theta}_{u+1}) - r_u^\star(\cdot; \bar{\theta}_u) \right\|_{\mathcal{H}}.$$

Assumption 4 (Bias–variance control under residual nonstationarity)**.** *The discrepancy estimator $\widehat{\delta}_{t-1}$ trained on data from $\tau, \ldots, t-1$ satisfies*

$$\mathbb{E}\Big[\|\widehat{\delta}_{t-1} - r_t^\star(\cdot; \bar{\theta}_t)\|_{\mathcal{H}}^2 \mid \mathcal{F}_{t-1} \Big] \leq \frac{A}{n_{\tau:t-1}} + B\Gamma_r(\tau, t-1)^2.$$

Assume also that within a locally homogeneous regime,

$$\left\| r_{u+1}^\star(\cdot; \bar{\theta}_{u+1}) - r_u^\star(\cdot; \bar{\theta}_u) \right\|_{\mathcal{H}} \leq \alpha_r.$$

Proposition 10 (Segment-length tradeoff for residual learning)**.** *If each batch contributes K observations and $L := t - 1 - \tau$ is the segment length in batches, then under Assumption 4,*

$$\mathbb{E}\Big[\|\widehat{\delta}_{t-1} - r_t^\star(\cdot; \bar{\theta}_t)\|_{\mathcal{H}}^2 \Big] \leq \frac{A}{KL} + B\alpha_r^2 L^2.$$

The minimizer satisfies

$$L^\star \asymp \left(\frac{A}{BK\alpha_r^2} \right)^{1/3}.$$

Proof. Under the local heterogeneity bound, $\Gamma_r(\tau, t-1) \leq \alpha_r L$. Substituting this into Assumption 4 yields the displayed upper bound. The stated scaling for $L^\star$ is obtained by differentiating $A/(KL) + B\alpha_r^2 L^2$ with respect to L and balancing the variance and bias terms. □

Interpretation. Longer segments reduce variance by providing more residual observations, but they increase bias when the residual target drifts because the simulator anchor changes over time. Restart therefore acts as adaptive data selection for discrepancy learning. This supports the design of BRPC-RRA: re-anchoring and refitting on segment-local residuals can improve BOCPD-facing evidence when abrupt segmentation quality is the main objective.

F.6 Auxiliary Results for wCUSUM and Re-Anchoring

The following statements support appendix-only analyses of the score-based controller and the residual re-anchoring construction. They are not part of the main conference theory narrative, but we record them here so that the proof section remains self-contained.

Proposition 11 (Idealized wCUSUM false-alarm bound)**.** *Let*

$$G_t = \max_{1 \le m \le W} \sqrt{m}\,(\bar{z}_{t-m+1:t} - \kappa)_+ .$$

Assume the oracle standardized score averages satisfy

$$\mathbb{P}\big(\sqrt{m}\,\bar{z}_{t-m+1:t} > h + \sqrt{m}\kappa\big) \le \exp\left(-\frac{(h+\sqrt{m}\kappa)^2}{2}\right)$$

for every relevant (t, m)*. Then*

$$\mathbb{P}\left(\max_{1 \le t \le T} G_t > h\right) \le \sum_{t=1}^{T}\sum_{m=1}^{W} \exp\left(-\frac{(h+\sqrt{m}\kappa)^2}{2}\right) \le TWe^{-h^2/2}.$$

In particular, $h \ge \sqrt{2\log(TW/\alpha)}$ *implies false-alarm probability at most* α*.*

Proposition 12 (Idealized wCUSUM detection guarantee)**.** *Suppose there exists a designated post-change block of length* $m_\star$ *on which the standardized score mean is shifted by at least* $\delta > \kappa$*, and that the sustained-shift concentration bound*

$$\mathbb{P}(\sqrt{m_\star}(\bar{z} - \delta) \le -u) \le e^{-u^2/2}$$

holds on that block. If

$$\sqrt{m_\star}(\delta - \kappa) - h \ge \sqrt{2\log(1/\beta)},$$

then the probability of failing to trigger restart within that block is at most β*. Equivalently, the required detection window scales like* $h^2/(\delta-\kappa)^2$ *up to logarithmic factors.*

Proposition 13 (Residual re-anchoring decoupling)**.** *For Exact, perturbing the previous discrepancy state by* Δ *changes the controller-facing discrepancy mean at time* t *by*

$$H_t A_t \Delta.$$

For BRPC-RRA, the posterior

$$q_{e,t}^{\mathrm{RRA}}(\delta) \propto p(\mathcal{D}_{e,t}^{\mathrm{res}} \mid \delta)\, p_0(\delta)$$

depends on the past only through the re-anchored residual dataset $\mathcal{D}_{e,t}^{\mathrm{res}}$*. Consequently, two previous discrepancy states that induce the same re-anchored residual dataset induce the same RRA posterior.*

Corollary 2 (Rate-optimality for persistent score shifts)**.** *In the idealized Gaussian mean-shift model, the preceding false-alarm and detection bounds imply the usual minimax-rate scaling for persistent score shifts. Suppose that before change the standardized scores are conditionally sub-Gaussian with mean at most* 0*, and after change there is a block of length* m *on which the standardized score mean is at least* $\delta > \kappa$*. Choose*

$$h = \sqrt{2\log(TW/\alpha)}.$$

Then Proposition 11 gives false-alarm probability at most α *over the horizon, and Proposition 12 implies detection within a window of length*

$$m \gtrsim \frac{\left(\sqrt{2\log(TW/\alpha)} + \sqrt{2\log(1/\beta)}\right)^2}{(\delta-\kappa)^2}$$

with probability at least $1-\beta$*, provided* $W \ge m$*.*

Proof. The rate in Corollary 2 is optimal up to logarithmic and drift-allowance factors in the standard Gaussian mean-shift model. Indeed, testing

$$H_0 : Z_1, \dots, Z_m \sim \mathcal{N}(0,1) \qquad \text{against} \qquad H_1 : Z_1, \dots, Z_m \sim \mathcal{N}(\delta,1)$$

has KL divergence $m\delta^2/2$. Any test with type-I error at most α and type-II error at most β must satisfy

$$\frac{m\delta^2}{2} \geq d(1-\beta \,\|\, \alpha),$$

where $d(p\|q)$ is the binary KL divergence. Hence

$$m \geq \frac{2\, d(1-\beta \,\|\, \alpha)}{\delta^2} = \Omega\left(\frac{\log(1/\alpha)}{\delta^2}\right)$$

for fixed detection probability. Therefore, in the persistent score-shift model, wCUSUM attains the optimal detection scaling up to logarithmic horizon/window factors and the drift allowance κ. □

G Complete Proofs

The BRPC-specific proofs most directly tied to the inner recursion are given earlier in Appendix E.2 and Appendix E.3. We continue here with the proofs for the additional BOCPD- and controller-facing theory statements. Appendix G.6 records an auxiliary contamination identity showing how parameter-anchor error perturbs the discrepancy loss; this result is not used in the main theorem chain but helps interpret residual-learning sensitivity.

G.1 Proof of the pre/post gap and pre-update bridge

Proof of Proposition 7. Let $x_t = a_t = A_t z_{t-1}$. From the first-order condition,

$$M_t(x_t - z_t) = \eta \nabla \ell_t(z_t). \tag{39}$$

Hence

$$\|x_t - z_t\|_{M_t} = \eta \, \|\nabla \ell_t(z_t)\|_{M_t^{-1}} \leq \eta G_t. \tag{40}$$

Since $\nabla^2 \ell_t = H_t^\top R_t^{-1} H_t$, the relative smoothness assumption implies

$$\nabla^2 \ell_t \preceq \frac{\rho_t}{\eta} M_t. \tag{41}$$

Therefore the quadratic upper model gives

$$\ell_t(x_t) \leq \ell_t(z_t) + \langle \nabla \ell_t(z_t), x_t - z_t \rangle + \frac{\rho_t}{2\eta} \|x_t - z_t\|_{M_t}^2 . \tag{42}$$

Using Cauchy–Schwarz and the norm bound,

$$\ell_t(x_t) - \ell_t(z_t) \leq \|\nabla \ell_t(z_t)\|_{M_t^{-1}} \|x_t - z_t\|_{M_t} + \frac{\rho_t}{2\eta} \|x_t - z_t\|_{M_t}^2 \tag{43}$$

$$\leq G_t(\eta G_t) + \frac{\rho_t}{2\eta}(\eta^2 G_t^2) = \eta \left(1 + \frac{\rho_t}{2}\right) G_t^2. \tag{44}$$

□

Proof of Corollary 1. Decompose

$$\ell_t(x_t) - \ell_t(u_t) = \big[\ell_t(x_t) - \ell_t(z_t)\big] + \big[\ell_t(z_t) - \ell_t(u_t)\big].$$

Apply Proposition 7 to the first term and Theorem 1 to the second, then sum over t. □

G.2 Proofs for the BOCPD mixture and combined surrogate theorem

Proof of Lemma 1. From the covariance decomposition,

$$\Sigma_{e,t}^{\mathrm{pre}} - R_{e,t} = \Sigma_{e,t}^{\mathrm{sim}} + H_{e,t} P_{e,t} H_{e,t}^{\top} \succeq 0.$$

Thus $\Sigma_{e,t}^{\mathrm{pre}} \succeq R_{e,t}$. Since both are positive definite, inverse order reverses:

$$(\Sigma_{e,t}^{\mathrm{pre}})^{-1} \preceq R_{e,t}^{-1}.$$

□

Proof of Proposition 8. By definition of the pre-update mean,

$$Y_t - \mu_{e,t}^{\mathrm{pre}} = \big(Y_t - \bar{\mu}_{e,t}(X_t)\big) - H_{e,t} x_{e,t} = r_{e,t} - H_{e,t} x_{e,t}. \tag{45}$$

By Lemma 1,

$$(\Sigma_{e,t}^{\mathrm{pre}})^{-1} \preceq R_{e,t}^{-1}.$$

Therefore,

$$\frac{1}{2} \left\| Y_t - \mu_{e,t}^{\mathrm{pre}} \right\|^2_{(\Sigma_{e,t}^{\mathrm{pre}})^{-1}} \leq \frac{1}{2} \left\| Y_t - \mu_{e,t}^{\mathrm{pre}} \right\|^2_{R_{e,t}^{-1}} \tag{46}$$

$$= \frac{1}{2} \left\| r_{e,t} - H_{e,t} x_{e,t} \right\|^2_{R_{e,t}^{-1}} = \ell_{e,t}^{\mathrm{prequad}}(x_{e,t}). \tag{47}$$

Adding the log-determinant and normalization terms proves the claim. □

Proof of Theorem 2. By definition,

$$\exp(-L_{1:T}^{\mathrm{mix}}) = \sum_{\pi' \in \Pi_T} \Pi(\pi') \exp\left[-\sum_{t=1}^{T} L_{e'_t,t}^{\mathrm{pre}} \right].$$

For any fixed path $\pi = (e_1, \ldots, e_T)$, one term in the sum is

$$\Pi(\pi) \exp\left[-\sum_{t=1}^{T} L_{e_t,t}^{\mathrm{pre}} \right].$$

Hence

$$\exp(-L_{1:T}^{\mathrm{mix}}) \geq \Pi(\pi) \exp\left[-\sum_{t=1}^{T} L_{e_t,t}^{\mathrm{pre}} \right].$$

Taking $-\log$ proves the result. □

Proof of Theorem 3. By Theorem 2,

$$L_{1:T}^{\mathrm{mix}} \leq \sum_{t=1}^{T} L_{e_t^\star,t}^{\mathrm{pre}} - \log \Pi(\pi^\star). \tag{48}$$

By Proposition 8,

$$L_{e_t^\star,t}^{\mathrm{pre}} \leq c_{e_t^\star,t} + \ell_{e_t^\star,t}^{\mathrm{prequad}}(x_t). \tag{49}$$

On each segment I_s, Corollary 1 yields

$$\sum_{t \in I_s} \left[\ell_{e_t^\star,t}^{\mathrm{prequad}}(x_t) - \ell_{e_t^\star,t}^{\mathrm{prequad}}(u_t^{(s)}) \right] \leq \frac{1}{2\eta} \left\| u_{\tau_{s-1}}^{(s)} - z_{\mathrm{init}} \right\|^2_{M_{\tau_{s-1}}} \tag{50}$$

$$+ \frac{1}{2\eta(1-\gamma)} \sum_{t \in I_s} \left\| u_t^{(s)} - A_t u_{t-1}^{(s)} \right\|^2_{M_t} + \sum_{t \in I_s} \Gamma_t. \tag{51}$$

Summing over segments and combining the three displays proves the theorem. □

G.3 Proof of Proposition 3

Proof of Proposition 3. Bayes updating with restart hazard h_t gives

$$\log \frac{w_t^{\text{restart}}}{w_t^{\text{cont}}} = \log \frac{h_t}{1-h_t} + \log \frac{p_n(y_t)}{p_c(y_t)}.$$

Expanding the Gaussian log-likelihood ratio yields the stated expression for $\Delta_t(y)$. Now suppose $Y_t \sim \mathcal{N}(\mu_t^\star, \Sigma_t^\star)$. For any Gaussian quadratic form,

$$\mathbb{E}[(Y-\mu)^\top \Sigma^{-1}(Y-\mu)] = \mathrm{Tr}(\Sigma^{-1}\Sigma_t^\star) + (\mu_t^\star - \mu)^\top \Sigma^{-1}(\mu_t^\star - \mu).$$

Applying this identity to continuation and fresh experts yields the claimed expression for $\mathbb{E}[\Delta_t(Y_t)]$. The final claim follows because when Σ_c is too small relative to $b_t = \mu_t^\star - \mu_c$, the term $b_t^\top \Sigma_c^{-1} b_t$ becomes large and can dominate the restart odds. □

G.4 Proofs for the idealized wCUSUM guarantees

Proof of Proposition 11. If $G_t > h$, then there exists some $m \in \{1, \dots, W\}$ such that

$$\sqrt{m}(\bar{z}_{t-m+1:t} - \kappa) > h,$$

equivalently,

$$\sqrt{m}\,\bar{z}_{t-m+1:t} > h + \sqrt{m}\kappa.$$

Therefore,

$$\mathbb{P}\left(\max_{1\le t\le T} G_t > h\right) \le \sum_{t=1}^{T}\sum_{m=1}^{W} \mathbb{P}\big(\sqrt{m}\,\bar{z}_{t-m+1:t} > h + \sqrt{m}\kappa\big) \tag{52}$$

$$\le \sum_{t=1}^{T}\sum_{m=1}^{W} \exp\left(-\frac{(h+\sqrt{m}\kappa)^2}{2}\right), \tag{53}$$

where the last step uses the oracle concentration assumption. Dropping the nonnegative $\sqrt{m}\kappa$ term gives the simpler upper bound $TWe^{-h^2/2}$. The choice $h \ge \sqrt{2\log(TW/\alpha)}$ then guarantees false-alarm probability at most α. □

Proof of Proposition 12. Consider the designated post-change block of length $m_\star$. On this block,

$$G_t \ge \sqrt{m_\star}(\bar{z} - \kappa)_+.$$

A failure to detect within the block implies

$$\sqrt{m_\star}(\bar{z} - \kappa) \le h,$$

or equivalently,

$$\sqrt{m_\star}(\bar{z} - \delta) \le h - \sqrt{m_\star}(\delta - \kappa).$$

Under the condition

$$\sqrt{m_\star}(\delta - \kappa) - h \ge \sqrt{2\log(1/\beta)},$$

we obtain

$$\mathbb{P}(\text{no detection in block}) \le \mathbb{P}\left(\sqrt{m_\star}(\bar{z} - \delta) \le -\sqrt{2\log(1/\beta)}\right) \le \beta,$$

using the sustained-shift concentration assumption. The delay scaling is the rearranged form of the same condition, up to logarithmic factors. □

G.5 Proof of the residual re-anchoring decoupling proposition

Proof of Proposition 13. For Exact, the pre-update state is $x_{e,t} = A_t z_{e,t-1}$. If the previous state is perturbed to $z_{e,t-1} + \Delta$, then

$$x'_{e,t} = A_t(z_{e,t-1} + \Delta) = x_{e,t} + A_t\Delta.$$

Applying the batch selection operator H_t yields the controller-facing discrepancy mean shift

$$H_t x'_{e,t} - H_t x_{e,t} = H_t A_t \Delta.$$

This proves the Exact part.

For BRPC-RRA, by definition

$$q_{e,t}^{\mathrm{RRA}}(\delta) \propto p(\mathcal{D}_{e,t}^{\mathrm{res}} \mid \delta) p_0(\delta).$$

Conditional on the re-anchored residual dataset $\mathcal{D}_{e,t}^{\mathrm{res}}$, the right-hand side contains no explicit dependence on the previous discrepancy state. Therefore, if two previous states induce the same residual dataset, they induce the same posterior. This proves the decoupling claim. □

G.6 Parameter-error contamination of discrepancy learning

Let $\bar{\mu}_t^\star(X_t)$ denote an oracle simulator anchor, and let $\bar{\mu}_t(X_t)$ denote the actual anchor. Define

$$\varepsilon_t^\theta := \bar{\mu}_t(X_t) - \bar{\mu}_t^\star(X_t). \tag{54}$$

Let

$$r_t^\star := Y_t - \bar{\mu}_t^\star(X_t), \qquad r_t := Y_t - \bar{\mu}_t(X_t), \tag{55}$$

so that

$$r_t = r_t^\star - \varepsilon_t^\theta. \tag{56}$$

For any state z, define

$$\ell_t(z; r) := \frac{1}{2} \|r - H_t z\|_{R_t^{-1}}^2 . \tag{57}$$

Proposition 14 (Exact contamination identity)**.** *For every t and every state z,*

$$\ell_t(z; r_t) = \ell_t(z; r_t^\star) - \left\langle \varepsilon_t^\theta, r_t^\star - H_t z \right\rangle_{R_t^{-1}} + \frac{1}{2} \left\|\varepsilon_t^\theta\right\|_{R_t^{-1}}^2 . \tag{58}$$

Proof. Substitute $r_t = r_t^\star - \varepsilon_t^\theta$ into the quadratic loss and expand. □

Corollary 3 (Pointwise perturbation bound)**.** *If*

$$\left\|\varepsilon_t^\theta\right\|_{R_t^{-1}} \le B_t^\theta, \tag{59}$$

then for every state z,

$$|\ell_t(z; r_t) - \ell_t(z; r_t^\star)| \le B_t^\theta \|r_t^\star - H_t z\|_{R_t^{-1}} + \frac{1}{2}(B_t^\theta)^2. \tag{60}$$

Proposition 15 (Regret perturbation identity)**.** *For every pair of states (z, u),*

$$\left[\ell_t(z; r_t) - \ell_t(u; r_t)\right] - \left[\ell_t(z; r_t^\star) - \ell_t(u; r_t^\star)\right] = \left\langle \varepsilon_t^\theta, H_t(z - u) \right\rangle_{R_t^{-1}} . \tag{61}$$

Proof. Apply the contamination identity to z and u and subtract. □

Corollary 4 (Quantitative contamination penalty)**.** *Assume*

$$\left\|\varepsilon_t^\theta\right\|_{R_t^{-1}} \le B_t^\theta, \qquad H_t^\top R_t^{-1} H_t \preceq \alpha_t M_t. \tag{62}$$

Then for every pair (z, u),

$$\left[\ell_t(z; r_t) - \ell_t(u; r_t)\right] \le \left[\ell_t(z; r_t^\star) - \ell_t(u; r_t^\star)\right] + \frac{\alpha_t}{2} \|z - u\|_{M_t}^2 + \frac{1}{2}(B_t^\theta)^2. \tag{63}$$

Proof. By the regret perturbation identity,

$$\left[\ell_t(z; r_t) - \ell_t(u; r_t)\right] - \left[\ell_t(z; r_t^\star) - \ell_t(u; r_t^\star)\right] = \left\langle \varepsilon_t^\theta, H_t(z - u) \right\rangle_{R_t^{-1}} .$$

Use Cauchy–Schwarz and the operator compatibility condition:

$$\left\langle \varepsilon_t^\theta, H_t(z - u) \right\rangle_{R_t^{-1}} \le \left\|\varepsilon_t^\theta\right\|_{R_t^{-1}} \|H_t(z - u)\|_{R_t^{-1}} \le B_t^\theta \sqrt{\alpha_t} \|z - u\|_{M_t} .$$

Now apply Young's inequality,

$$ab \le \frac{1}{2} a^2 + \frac{1}{2} b^2,$$

with $a = B_t^\theta$ and $b = \sqrt{\alpha_t} \|z - u\|_{M_t}$. □

Coupled interpretation with projected-PF error. The main tracking theorem is conditional on the residual sequence supplied to the discrepancy learner. The projected particle filter enters through the residual anchor. Let

$$r_t^\star = Y_t - y_s(X_t, \theta_t^\star), \qquad r_t = Y_t - \bar{\mu}_t(X_t),$$

where $\theta_t^\star$ is an oracle projected calibration target and $\bar{\mu}_t(X_t)$ is the simulator anchor produced by the projected PF. Define the anchor error

$$e_t^\theta := \bar{\mu}_t(X_t) - y_s(X_t, \theta_t^\star), \qquad r_t = r_t^\star - e_t^\theta.$$

Thus PF error enters the discrepancy learner only through the residual contamination e_t^θ.

For the quadratic residual loss

$$\ell_t(z; r) = \frac{1}{2}\|r - G_t z\|^2_{R_t^{-1}},$$

Proposition 14 gives

$$\ell_t(z; r_t) - \ell_t(v_t; r_t) = \ell_t(z; r_t^\star) - \ell_t(v_t; r_t^\star) + \langle e_t^\theta, G_t(z - v_t)\rangle_{R_t^{-1}}.$$

Consequently, if $\|e_t^\theta\|_{R_t^{-1}} \le B_t^\theta$ and $G_t^\top R_t^{-1} G_t \preceq \alpha_t M_t$, then

$$\ell_t(z; r_t^\star) - \ell_t(v_t; r_t^\star) \le \ell_t(z; r_t) - \ell_t(v_t; r_t) + \frac{\alpha_t}{2}\|z - v_t\|^2_{M_t} + \frac{1}{2}(B_t^\theta)^2.$$

Combining this inequality with Theorem 1 shows that the end-to-end discrepancy tracking error consists of the pathwise BRPC tracking terms plus an additive penalty controlled by the projected-PF anchor error. When the projected PF keeps the simulator anchor close to the oracle projected target, this penalty is small; when the PF anchor is biased or poorly covered by the particles, discrepancy learning is contaminated through the residuals.

G.7 One-Step Contraction of the BRPC-E Update

The transport condition used in the main text can be paired with a direct nonexpansiveness property of the BRPC-E proximal update. Fix a batch (X_t, Y_t) and write

$$G_t := H_t^\top R_t^{-1} H_t.$$

For a predictive prior mean a, define the BRPC-E update map

$$T_t(a) := \arg\min_z \left\{ \frac{1}{2}\|z - a\|^2_{M_t} + \eta \frac{1}{2}\|r_t - H_t z\|^2_{R_t^{-1}} \right\}.$$

Proposition 16 (One-step nonexpansiveness)**.** *For any two predictive prior means a, a',*

$$\|T_t(a) - T_t(a')\|_{M_t} \le \|a - a'\|_{M_t}.$$

If, on a subspace $\mathcal{U}$, the likelihood curvature satisfies

$$G_t \succeq \mu_t M_t \qquad \text{on } \mathcal{U}$$

for some $\mu_t > 0$, then for differences $a - a' \in \mathcal{U}$,

$$\|T_t(a) - T_t(a')\|_{M_t} \le \frac{1}{1 + \eta\mu_t}\|a - a'\|_{M_t}.$$

Proof. The closed-form update is

$$T_t(a) = (M_t + \eta G_t)^{-1}\left(M_t a + \eta H_t^\top R_t^{-1} r_t\right).$$

Thus

$$T_t(a) - T_t(a') = (M_t + \eta G_t)^{-1} M_t (a - a').$$

Let

$$B_t := M_t^{-1/2} G_t M_t^{-1/2} \succeq 0.$$

Then

$$M_t^{1/2}\{T_t(a) - T_t(a')\} = (I + \eta B_t)^{-1} M_t^{1/2}(a - a').$$

Since $B_t \succeq 0$, all eigenvalues of $(I + \eta B_t)^{-1}$ lie in $(0, 1]$, which gives nonexpansiveness in the M_t norm. If $G_t \succeq \mu_t M_t$ on $\mathcal{U}$, then $B_t \succeq \mu_t I$ on the corresponding transformed subspace, so

$$\|(I + \eta B_t)^{-1}\| \le \frac{1}{1 + \eta\mu_t},$$

which gives the strict contraction bound. □

Corollary 5 (Transport-plus-update contraction)**.** *Suppose the predictive prior mean is transported as $a_t = A_t z_{t-1}$ and Assumption 1 holds:*

$$A_t^\top M_t A_t \preceq \gamma M_{t-1}.$$

For two BRPC-E trajectories updated on the same residual batch,

$$\|z_t - z'_t\|_{M_t} \le \sqrt{\gamma}\|z_{t-1} - z'_{t-1}\|_{M_{t-1}}.$$

If, additionally, $G_t \succeq \mu_t M_t$ on the relevant difference directions, then

$$\|z_t - z'_t\|_{M_t} \le \frac{\sqrt{\gamma}}{1+\eta\mu_t}\|z_{t-1} - z'_{t-1}\|_{M_{t-1}}.$$

Proof. Apply the previous proposition with $a_t = A_t z_{t-1}$ and $a'_t = A_t z'_{t-1}$. Then

$$\|T_t(a_t) - T_t(a'_t)\|_{M_t} \le \|A_t(z_{t-1} - z'_{t-1})\|_{M_t}.$$

Assumption 1 gives

$$\|A_t(z_{t-1} - z'_{t-1})\|_{M_t}^2 \le \gamma\|z_{t-1} - z'_{t-1}\|_{M_{t-1}}^2.$$

Taking square roots proves the first claim. The strict version follows by combining the same transport bound with the strict contraction factor from the one-step update. □

Why restart helps under jumps. Theorem 1 also clarifies why an outer controller is needed under abrupt regime change. Without restart, a single BRPC-E trajectory competes with one transported comparator path and pays the full variation cost

$$\sum_{t=1}^{T} \|u_t - A_t u_{t-1}\|_{M_t}^2.$$

If the target has abrupt jumps, this term contains large jump contributions. With restart times $0 = \tau_0 < \tau_1 < \cdots < \tau_S = T$, the same argument applied segmentwise gives

$$\sum_{s=1}^{S}\sum_{t=\tau_{s-1}+1}^{\tau_s}\{\ell_t(z_t^{(s)}) - \ell_t(u_t^{(s)})\} \le \sum_{s=1}^{S}\frac{1}{2\eta}\|u_{\tau_{s-1}}^{(s)} - z_{\text{init}}\|_{M_{\tau_{s-1}}}^2 + \sum_{s=1}^{S}\frac{1}{2\eta(1-\gamma)}\sum_{t=\tau_{s-1}+1}^{\tau_s}\|u_t^{(s)} - A_t u_{t-1}^{(s)}\|_{M_t}^2.$$

When restarts align with large jumps, the within-segment variation terms exclude the jump discontinuities and are therefore much smaller. This is the formal reason that a stable local BRPC-E recursion and an explicit outer restart controller play complementary roles: contraction stabilizes within-regime learning, while restart prevents abrupt regime changes from being treated as large transported-memory errors.

H Benchmark Descriptions

I Benchmark Descriptions

I.1 Synthetic Benchmark

The synthetic benchmark adapts the static calibration example of Gu and Wang [6] to an online nonstationary stream. The simulator is

$$y_s(x, \theta) = \sin(\theta x) + 5x,$$

and the physical response is

$$\zeta_t(x) = 5x\cos(\omega_t x/2) + 5x,$$

observed with independent Gaussian noise,

$$Y_t(x) = \zeta_t(x) + \epsilon_t(x), \qquad \epsilon_t(x) \sim \mathcal{N}(0, \sigma_y^2).$$

In the original static example, $\omega_t = 15$ is fixed. We turn this example into an online benchmark by allowing ω_t and the corresponding projected calibration target to drift smoothly, jump abruptly,

Table 2: Synthetic benchmark configurations. Segment length denotes the number of observations between consecutive annotated changepoints in sudden-jump streams. The mixed family combines the drifting and sudden-jump mechanisms.

Setting	Configuration
Sudden-jump family	
Segment length (L)	$\{80, 120, 200\}$
Jump magnitude	$\{0.5, 1.0, 2.0, 3.0\}$
Batch size (K_t)	$\{10, 20, 40\}$
Total observations	$4L$
Particles	1024
Experts maintained	5
Drifting family	
Drift slope (s)	$\{5\times10^{-4}, 10^{-3}, 1.5\times10^{-3}, 2\times10^{-3}, 2.5\times10^{-3}\}$
Batch size (K_t)	$\{10, 20, 40\}$
Total observations	600
Particles	1024
Experts maintained	5
Mixed family	
Drift component	Same slope-style drift family as above
Jump component	Annotated abrupt changes with varying magnitude
Batch size (K_t)	$\{10, 20, 40\}$
Particles	1024
Experts maintained	5

or combine both behaviors. This produces simulator–reality mismatch while retaining a known nonstationary structure.

We use three scenario families. In the drifting family, the latent regime variable evolves smoothly, so the main challenge is local adaptation rather than explicit segmentation. This family includes random-walk and slope-style trajectories of increasing difficulty. In the sudden family, the latent regime variable is piecewise constant with abrupt jumps at unknown times, so the main challenge is fast restart without excessive false alarms. In the mixed family, smooth drift and abrupt jumps coexist, producing the compatibility stress test used throughout the paper. Across these families, we vary drift strength, jump magnitude, segment length, and batch size. Event-level restart metrics such as Precision@2, Recall@2, and Delay@2 are computed only for settings with annotated abrupt changes, namely the sudden and mixed families.

Unless otherwise stated, the number of particles is fixed at 1024 for particle-based methods, and the maximum number of BOCPD experts maintained online is fixed at 5. The configuration ranges used in the synthetic study are summarized in Table 2.

I.2 Plant-Simulation Benchmark

The plant-simulation benchmark complements the stylized synthetic study with a more realistic digital-twin synchronization task. The physical system is a bicycle-production line implemented in Plant Simulation, with stochastic arrivals and processing times, work-in-progress and inventory constraints, and regime-dependent demand. The latent parameter to be tracked is an arrival-rate variable that determines the effective system load. Figure 2 shows the digital-twin environment used for this case study.

The simulated line contains three main stations (Assembly, Cleaning, and Inspection), two buffers (CustomerBuffer and KitBuffer), and an (R, Q) inventory policy. Each batch also carries an exogenous control vector

$$X = (M_1, M_2, W, R, Q),$$

where M_1 and M_2 are station-level factor settings, W is a workload factor, and (R, Q) are inventory controls. The control ranges are summarized in Table 3.

We generate 2,000 Plant Simulation runs by Latin hypercube sampling over the control space and train a surrogate model for 10-day net revenue, which becomes the streaming response used by the

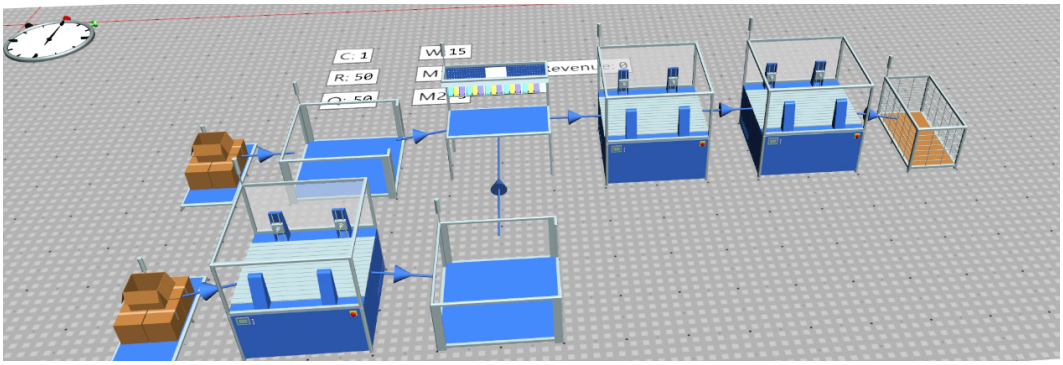


Figure 2: Digital-twin environment for the bicycle-production plant-simulation benchmark. The online calibration stream is generated from this Plant Simulation model under multiple operating modes.

Table 3: Control factors used in the bicycle-plant simulation benchmark.

Factor	Levels / Range	Description
M_1	$\{1, 2, 3\}$	Assembly-related setting
M_2	$\{1, 2, 3\}$	Cleaning/inspection-related setting
W	$[10, 15]$	Workload factor
R	$[50, 100]$	Reorder point
Q	$[50, 100]$	Order quantity

online calibrator. Preliminary experiments found that a standard Gaussian-process surrogate underfit this response surface, so the benchmark uses a four-layer multilayer perceptron surrogate instead.

For the physical data stream, we sequentially sample control inputs and generate observations under three representative operating modes:

1. **Drifting mode:** the latent arrival-rate parameter varies smoothly. In the sinusoidal configuration,
$$\theta_t = 11.5 + 8.5 \sin\left(\frac{2\pi t}{40}\right).$$
2. **Sudden-change mode:** the latent arrival-rate parameter is piecewise constant with abrupt regime switches.
3. **Mixed mode:** the latent arrival-rate parameter combines continuous variation with abrupt structural changes.

The goal is to infer the latent arrival-rate parameter online from the revenue stream while maintaining synchronization between the physical system and the digital twin. This benchmark is intended to test whether the qualitative conclusions from the synthetic study persist when parameter mismatch, surrogate error, and operational stochasticity are entangled in a realistic simulator. Reported relative y-metrics are normalized by the mode-specific response scale so that predictive performance is comparable across operating modes.

J Additional Experiment Results

J.1 Transport-Metric Replay Diagnostic

Assumption 1 is a sufficient pre-update transport condition used in the tracking analysis. It should not be interpreted as a claim that every realized posterior mean update is contractive. To inspect the corresponding stability geometry in representative streams, we compute two replay diagnostics from saved or regenerated input batches.

Table 4: Transport-metric replay diagnostic on representative mixed streams. γ_{prior} is the pre-update transport diagnostic closest to Assumption 1. γ_{post} uses the post-update precision and therefore measures posterior sharpening rather than transport stability.

Dataset	Scenario	n	Median γ_{prior}	Max γ_{prior}	Frac. $\gamma_{\text{prior}} \leq 1$	Median γ_{post}	Max γ_{post}
Synthetic	Mixed	145	0.9999	1.0000	0.9517	1.0666	1.9844
PlantSim	Mixed	1060	0.1532	0.9877	1.0000	0.5846	2.1394

Let P_t^- denote the transported pre-update covariance of the discrepancy state and let C_t denote the post-update covariance after assimilating the current residual batch. We compute

$$\gamma_{\text{prior},t} = \lambda_{\max}\left(M_{t-1}^{-1/2} A_t^\top (P_t^-)^{-1} A_t M_{t-1}^{-1/2}\right),$$

where $M_{t-1} = C_{t-1}^{-1}$. This is the closest empirical analogue of Assumption 1 because it uses the transported pre-update precision. We also compute

$$\gamma_{\text{post},t} = \lambda_{\max}\left(M_{t-1}^{-1/2} A_t^\top C_t^{-1} A_t M_{t-1}^{-1/2}\right).$$

The second quantity is not an Assumption 1 diagnostic. It measures posterior sharpening after the current batch is assimilated. Since

$$C_t^{-1} = (P_t^-)^{-1} + \eta_\delta G_t^\top R_t^{-1} G_t,$$

$\gamma_{\text{post},t}$ can exceed one even when the pre-update transport is stable.

The replay uses the same input batches as the experiment streams. For the synthetic benchmark, we deduplicate method labels and replay each saved design once, because this diagnostic depends on input-batch geometry and kernel/noise settings rather than on the BRPC controller label. For the plant-simulation benchmark, we regenerate the input batches for the mixed operating mode and random seeds. The likelihood tempering is set by $\eta_\delta = 1/\lambda_\delta$, so the effective residual covariance in the Gaussian covariance update is R/η_δ.

Table 4 reports the replay diagnostic on the mixed synthetic and plant-simulation streams, which contain both gradual drift and abrupt changes. The pre-update diagnostic γ_{prior} is nearly nonexpansive on the synthetic stream and strictly contractive on the plant-simulation stream. Thus, in this representative diagnostic, the transport-metric analogue of Assumption 1 is empirically stable.

The post-update diagnostic behaves differently. In both benchmarks, γ_{post} can be larger than one, especially in the upper tail. This is expected because the current residual batch increases posterior precision through the Gaussian update. These values should therefore not be read as failures of Assumption 1. They indicate posterior sharpening after data assimilation, whereas Assumption 1 concerns the pre-update transport geometry.

This diagnostic is covariance-level rather than response-value-level. Under the fixed Gaussian replay used here, the covariance recursion depends on the input batches, kernel hyperparameters, residual noise, and likelihood tempering, but not on the realized response values. The response values affect the posterior mean and the realized discrepancy update, while the replay diagnostic isolates the transport metric used in the sufficient stability condition.

J.2 Joint EnKF Sensitivity Study

This section reports a sensitivity study for a joint ensemble Kalman filtering baseline. The purpose is not to introduce a new method, but to check whether a direct joint-assimilation alternative to BRPC can be made competitive by tuning its process-noise and inflation parameters.

Joint state and observation model. The JointEnKF baseline maintains an ensemble over a joint calibration–bias state

$$z_t = \left(\theta_t, \beta_t^\top\right)^\top, \qquad \beta_t \in \mathbb{R}^q.$$

Here θ_t is the calibration parameter and β_t is a finite discrepancy-coefficient vector. Given basis functions $b(x) \in \mathbb{R}^q$, the observation model used by the filter is

$$Y_t(x) = y_s(x, \theta_t) + b(x)^\top \beta_t + \epsilon_t, \qquad \epsilon_t \sim N(0, \sigma^2).$$

For the one-dimensional synthetic benchmark, we use an intercept, a linear term, and radial-basis functions,

$$b(x) = \left(1,\ x,\ \exp\left[-\frac{(x-c_1)^2}{2\ell_b^2}\right], \ldots, \exp\left[-\frac{(x-c_m)^2}{2\ell_b^2}\right]\right)^\top.$$

Thus the baseline is genuinely joint: the calibration parameter and the discrepancy coefficients are updated together from the same batch innovation.

The forecast step evolves both components by damped random walk:

$$\theta_t^{(i),-} = \Pi_\Theta\left(\theta_{t-1}^{(i)} + \xi_{\theta,t}^{(i)}\right), \qquad \xi_{\theta,t}^{(i)} \sim N(0, \sigma_\theta^2),$$

and

$$\beta_t^{(i),-} = \rho_\beta \beta_{t-1}^{(i)} + \xi_{\beta,t}^{(i)}, \qquad \xi_{\beta,t}^{(i)} \sim N(0, \sigma_\beta^2 I_q),$$

where Π_Θ clips θ to the admissible range.

Given a batch X_t, each ensemble member induces a predicted observation vector

$$\widehat{Y}_t^{(i),-} = y_s(X_t, \theta_t^{(i),-}) + B_t \beta_t^{(i),-}, \qquad B_t = \begin{bmatrix} b(x_{t,1})^\top \\ \vdots \\ b(x_{t,K_t})^\top \end{bmatrix}.$$

Let Z_t^- be the matrix of forecast joint states and let $\widehat{Y}_t^-$ be the matrix of predicted batch observations. The EnKF uses empirical cross-covariance and observation covariance

$$C_{zy} = \frac{1}{N-1}\widetilde{Z}_t^\top \widetilde{Y}_t, \qquad C_{yy} = \frac{1}{N-1}\widetilde{Y}_t^\top \widetilde{Y}_t + \sigma^2 I,$$

where $\widetilde{Z}_t$ and $\widetilde{Y}_t$ are the ensemble anomaly matrices, with covariance inflation applied to the state anomalies. The Kalman gain is

$$K_t = C_{zy} C_{yy}^{-1}.$$

Using perturbed observations, each ensemble member is updated as

$$z_t^{(i)} = z_t^{(i),-} + K_t\left(Y_t + \epsilon_t^{(i)} - \widehat{Y}_t^{(i),-}\right), \qquad \epsilon_t^{(i)} \sim N(0, \sigma^2 I).$$

This update directly contrasts with BRPC: in JointEnKF, θ_t and β_t are corrected through the same innovation, whereas BRPC updates θ_t through a discrepancy-free projected likelihood and learns discrepancy only conditionally afterward.

Sensitivity grid. To check that the JointEnKF baseline was not disadvantaged by an arbitrary hyperparameter choice, we conducted a sensitivity study over the main state-evolution parameters used by the filter. Specifically, we varied the random-walk standard deviation for the calibration parameter state, the random-walk standard deviation for the discrepancy-coefficient state, and the covariance inflation factor:

$$\sigma_\theta \in \{0.015,\ 0.035,\ 0.07\}, \qquad \sigma_\beta \in \{0.005,\ 0.015,\ 0.04\}, \qquad \kappa_{\text{infl}} \in \{1.00,\ 1.02,\ 1.08\}.$$

For each hyperparameter combination, we ran JointEnKF on the synthetic SLOPE, SUDDEN, and MIXED scenarios and recorded θ-RMSE and y-RMSE.

Figure 3 shows the resulting θ-RMSE sensitivity heatmaps. To keep the figure readable, we do not print the numeric values inside each cell. Instead, the color indicates the relative performance of that hyperparameter combination within the same scenario.

The main conclusion is that JointEnKF is sensitive to the allocation of process noise and inflation. The sensitivity landscapes are not flat, and no single hyperparameter choice dominates uniformly across all nonstationary scenarios. This supports two points. First, the reported JointEnKF behavior is not the result of selecting an artificially poor configuration. Second, joint assimilation itself requires substantial tuning to balance fast adaptation against filter instability and parameter–bias attribution.

We emphasize that this sensitivity analysis is included as a robustness check for the baseline rather than as a central contribution of the paper. Our main comparison remains focused on projected online calibration with and without explicit outer control.

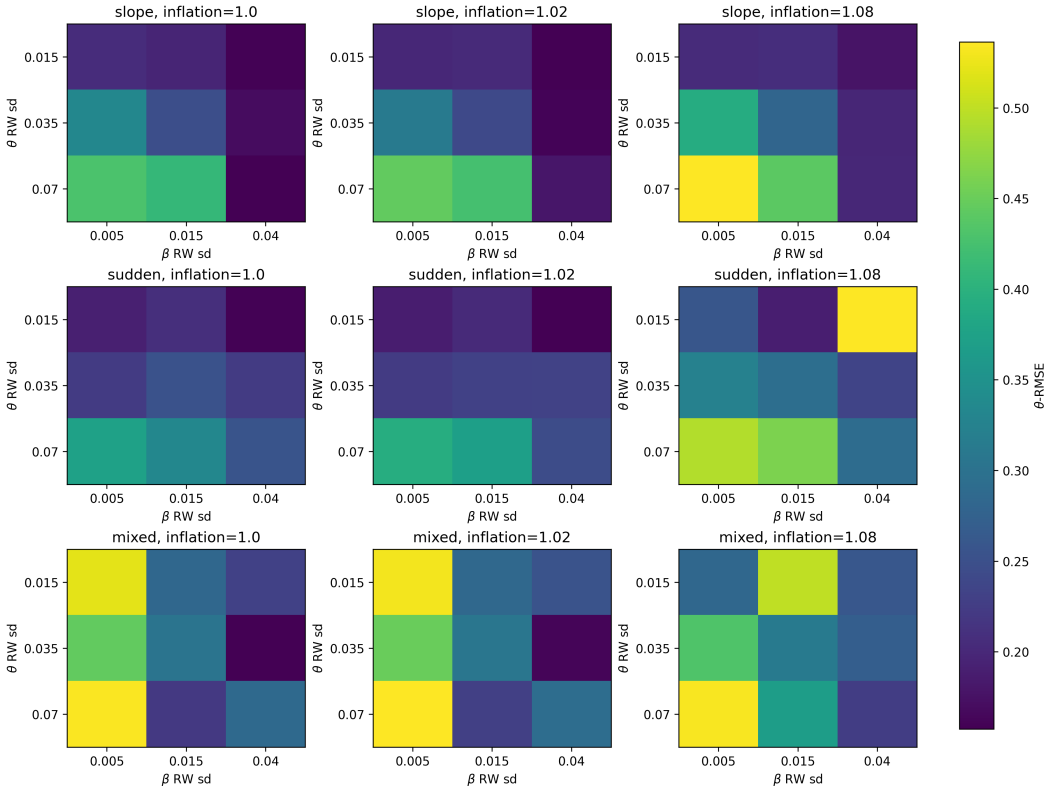


Figure 3: JointEnKF sensitivity heatmaps for θ-RMSE on the synthetic SLOPE, SUDDEN, and MIXED scenarios. Columns correspond to covariance inflation settings, rows correspond to scenarios, and each heatmap cell corresponds to one $(\sigma_\theta, \sigma_\beta)$ combination. Darker cells indicate lower θ-RMSE within the same scenario. The figure shows that JointEnKF performance depends materially on process-noise allocation and inflation.

J.3 Supplementary BRPC-P and BRPC-F Ablations

In the headline tables, **B-BRPC** and **C-BRPC** abbreviate the default Exact recursive implementation, i.e., **B-BRPC-E** and **C-BRPC-E**. This appendix expands the implementation-level comparison to proxy-memory and fixed-support variants.

The main text reports the default Exact implementation together with **B-BRPC-RRA**. Tables 5 and 6 report fuller ablations for **BRPC-P** and **BRPC-F**. These results test whether the controller-side conclusions persist after replacing the Exact discrepancy state by proxy memory or a fixed-support projection.

On the synthetic benchmark, **BRPC-F** closely mirrors the Exact family. In particular, **B-BRPC-F** and **C-BRPC-F** reproduce nearly the same restart counts and predictive behavior as **B-BRPC-E** and **C-BRPC-E**, which is consistent with the view that the fixed-support approximation preserves the controller-facing evidence geometry while only changing the internal state representation. **BRPC-P** is more mode-dependent. Uncontrolled **BRPC-P** remains clearly weaker than the restarted variants, but **B-BRPC-P** and **C-BRPC-P** can be competitive, sometimes with lower restart burden than the Exact family. The tradeoff is that proxy memory is typically slower and its gains are less uniform across regimes. Overall, the synthetic appendix results support the same qualitative picture as the headline tables: controller choice continues to matter more than the precise recursive approximation.

The plant-simulation ablation in Table 6 reinforces the same message. Fixed-support variants again track the Exact controller pattern closely: BOCPD-controlled versions restart more often than their wCUSUM-controlled counterparts, while predictive differences remain secondary. Proxy-memory variants can be competitive, especially in the mixed and drifting modes, but they do not produce a qualitatively different conclusion. Most importantly, none of these auxiliary variants eliminate the

Table 5: Synthetic benchmark results across the *drifting*, *sudden*, and *mixed* scenario families. We report mean performance over repeated runs. Lower is better for θ-RMSE, θ-CRPS, y-RMSE, y-CRPS, and runtime. Restart count is reported as a controller-behavior diagnostic.

Scenario	Method	θ-RMSE ↓	θ-CRPS ↓	y-RMSE ↓	y-CRPS ↓	Restarts	Time (s) ↓
Drifting	DA(WaldPF)	0.090	0.064	1.909	1.090	–	1.279
	DA(EnKF)	0.109	0.066	0.548	0.295	–	0.236
	B-BRPC-E	0.014	0.025	0.484	0.302	11.774	39.402
	C-BRPC-E	0.015	0.024	0.580	0.375	4.774	19.238
	BRPC-F	0.015	0.024	1.160	0.764	–	15.662
	B-BRPC-F	0.015	0.025	0.484	0.302	11.758	37.490
	C-BRPC-F	0.014	0.024	0.581	0.376	4.774	16.785
	BRPC-P	0.015	0.024	0.728	0.418	–	75.991
	B-BRPC-P	**0.014**	0.032	**0.408**	0.247	4.440	88.171
	C-BRPC-P	0.015	0.024	0.558	0.334	1.782	40.775
	B-BRPC-RRA	0.015	0.039	0.437	0.240	8.128	70.667
Sudden(3)	DA(WaldPF)	0.178	0.110	1.994	1.157	–	1.160
	DA(EnKF)	0.160	0.090	0.982	0.600	–	0.189
	B-BRPC-E	0.026	0.029	0.607	0.400	10.287	32.020
	C-BRPC-E	0.027	0.027	0.671	0.445	3.706	16.051
	BRPC-F	0.040	0.031	1.192	0.791	–	12.847
	B-BRPC-F	0.026	0.029	0.609	0.401	10.322	30.414
	C-BRPC-F	0.027	0.027	0.673	0.446	3.746	13.858
	BRPC-P	0.041	0.031	0.867	0.515	–	59.710
	B-BRPC-P	0.028	0.034	0.590	0.375	4.054	73.600
	C-BRPC-P	0.026	0.027	0.577	0.367	2.784	27.000
	B-BRPC-RRA	**0.018**	0.028	0.512	0.316	3.050	93.857
Mixed(3)	DA(WaldPF)	0.092	0.062	1.873	1.077	–	1.279
	DA(EnKF)	0.346	0.065	0.653	0.394	–	0.239
	B-BRPC-E	0.021	0.027	0.464	0.292	11.865	32.857
	C-BRPC-E	0.020	0.025	0.535	0.336	3.493	16.790
	BRPC-F	0.030	0.027	1.032	0.665	–	12.513
	B-BRPC-F	0.022	0.027	0.471	0.298	11.867	30.606
	C-BRPC-F	0.021	0.025	0.536	0.337	3.493	13.515
	BRPC-P	0.031	0.028	0.791	0.467	–	64.380
	B-BRPC-P	0.021	0.032	0.499	0.309	4.616	73.571
	C-BRPC-P	**0.016**	0.025	0.494	0.307	3.041	25.215
	B-BRPC-RRA	0.021	0.036	0.505	0.299	4.813	78.127

asymmetry highlighted in the main text: controller-side fixes are the most robust way to reduce restart burden across benchmarks, while residual re-anchoring is most compelling in jump-heavy settings.

Scalability. The main computational costs separate into parameter filtering and discrepancy learning. With N particles and calibration dimension d_θ, the projected parameter update scales with the cost of propagating and scoring the particles, typically $O(N)$ simulator or surrogate evaluations per batch, plus resampling. The discrepancy update is controlled by the chosen BRPC variant. BRPC-E has an expanding support and is appropriate for moderate segment lengths; BRPC-F uses a fixed support of size M and has update cost dominated by fixed-rank linear algebra, e.g. $O(M^3)$ initialization and $O(B_t M^2)$ batch updates under the standard fixed-support representation. RRA is heavier because it refits on the current segment; with n_e archived observations, an exact refit costs $O(n_e^3)$, while a fixed-rank version costs $O(n_e M^2 + M^3)$. Thus, for larger input spaces or longer streams, the scalable implementation is BRPC-F with score-based reset control, whereas RRA is best reserved for shorter, jump-dominated segments where segment-local residual evidence is worth the extra cost.

J.4 Controller and State-Sharing Ablations

This section studies two auxiliary questions that are not central to the main headline tables but help clarify the design choices in BRPC. First, we ask whether an explicit outer restart controller is necessary, by comparing fixed-support discrepancy learning with no controller, BOCPD restart

Table 6: Additional plant-simulation ablation results across the three operating modes. We report mean performance over repeated runs. Lower is better for θ-RMSE, θ-CRPS, relative y-RMSE, relative y-CRPS, and runtime. Here, relative y-RMSE and relative y-CRPS denote response errors normalized by the mode-specific response scale, so the reported y metrics are dimensionless and comparable across modes. Restart count is reported as a controller-behavior diagnostic.

Mode	Method	θ-RMSE ↓	θ-CRPS ↓	Rel. y-RMSE ↓	Rel. y-CRPS ↓	Restarts	Time (s) ↓
Drifting	B-BRPC-E	0.800	0.454	0.988	0.274	15.700	46.714
	C-BRPC-E	0.809	0.451	1.001	0.280	7.000	24.304
	B-BRPC-F	0.801	0.457	0.980	0.274	15.900	35.584
	C-BRPC-F	0.807	0.451	1.020	0.292	5.900	17.438
	B-BRPC-RRA	0.788	0.444	1.109	0.302	17.900	93.203
	B-BRPC-P	0.802	0.454	0.960	0.256	7.100	113.778
	C-BRPC-P	0.813	0.452	0.991	0.262	6.000	46.636
Sudden(5)	B-BRPC-E	0.957	0.509	0.164	0.079	7.400	57.009
	C-BRPC-E	1.056	0.531	0.169	0.083	4.300	27.513
	B-BRPC-F	1.141	0.596	0.177	0.085	7.000	37.952
	C-BRPC-F	1.020	0.522	0.179	0.085	2.900	16.444
	B-BRPC-RRA	1.048	0.531	0.128	0.047	5.000	149.955
	B-BRPC-P	1.309	0.683	0.163	0.087	7.400	118.882
	C-BRPC-P	1.024	0.519	0.171	0.084	3.500	67.723
Mixed	B-BRPC-E	1.580	0.673	0.554	0.174	12.400	34.237
	C-BRPC-E	1.582	0.704	0.561	0.179	5.700	18.143
	B-BRPC-F	1.588	0.736	0.555	0.193	12.000	27.276
	C-BRPC-F	1.527	0.785	0.593	0.182	5.600	13.499
	B-BRPC-RRA	1.537	0.655	0.569	0.178	13.000	71.213
	B-BRPC-P	1.545	0.719	0.557	0.186	8.100	69.558
	C-BRPC-P	1.564	0.692	0.575	0.184	5.300	33.486

control, and wCUSUM restart control. Second, we ask whether particle-specific discrepancy states provide a practical advantage over the shared expert-level discrepancy state used in the main paper.

Table 7 reports the comparison for the shared fixed-support variant and its particle-specific analogue. The "None" rows use the same local fixed-support discrepancy learner but do not perform explicit outer restart. The BOCPD and wCUSUM rows use the corresponding restart controllers described in the main text.

The first pattern is that the outer controller is important whenever abrupt changes or mixed non-stationarity are present. In the sudden family, the shared fixed-support learner without a controller has y-RMSE 1.1915, whereas BOCPD and wCUSUM reduce it to 0.6089 and 0.6727, respectively. The particle-specific variant shows the same behavior: y-RMSE decreases from 1.2092 without a controller to 0.6504 under BOCPD and 0.6857 under wCUSUM. In the mixed family, the same gap is visible: the shared fixed-support learner has y-RMSE 1.0319 without a controller, compared with 0.4714 under BOCPD and 0.5357 under wCUSUM. These results support the main design premise that local discrepancy adaptation alone is not enough under regime change; an outer controller is needed to manage stale memory and reset the calibration state.

The second pattern is that particle-specific fixed-support discrepancy states do not provide a systematic advantage over the shared expert-level fixed-support state. Across controller choices and scenario families, the two variants have very similar θ-RMSE. However, the particle-specific version is usually slightly worse in predictive error and is consistently slower. For example, in the mixed family under BOCPD, shared fixed support obtains y-RMSE 0.4714 in 30.606 seconds, whereas particle-specific fixed support obtains 0.4959 in 39.409 seconds. In the sudden family under BOCPD, the corresponding values are 0.6089 versus 0.6504 for y-RMSE and 30.414 versus 36.592 seconds for runtime. The wCUSUM rows show the same qualitative pattern.

These ablations justify the shared expert-level discrepancy design used in the main experiments. Particle-specific discrepancy states are conceptually closer to the fully conditional projected-calibration construction, but in this benchmark they do not improve tracking or prediction enough to offset their higher variance and runtime. The shared state therefore appears to be a better practical compromise: it preserves the projected residual logic while producing a simpler and more stable controller-facing evidence stream.

Table 7: Controller and state-sharing ablation for BRPC-F learning on the synthetic benchmark. Values are mean ± standard deviation across runs. Restart count is a controller-behavior diagnostic rather than a monotone performance metric.

Scenario	State	Controller	θ-RMSE ↓	y-RMSE ↓	Restarts	Time (s) ↓
Drifting	Shared	None	0.0145 ± 0.0021	1.1603 ± 0.3362	0.000 ± 0.000	15.662
	Shared	BOCPD	0.0146 ± 0.0021	0.4839 ± 0.2251	11.758 ± 1.755	37.490
	Shared	wCUSUM	0.0145 ± 0.0022	0.5809 ± 0.2628	4.774 ± 0.420	16.785
	Particle	None	0.0145 ± 0.0021	1.1881 ± 0.3370	0.000 ± 0.000	18.100
	Particle	BOCPD	0.0145 ± 0.0021	0.5141 ± 0.2451	11.260 ± 1.639	42.830
	Particle	wCUSUM	0.0144 ± 0.0022	0.6040 ± 0.2703	4.774 ± 0.420	19.567
Sudden	Shared	None	0.0401 ± 0.0342	1.1915 ± 0.3850	0.000 ± 0.000	12.847
	Shared	BOCPD	0.0264 ± 0.0235	0.6089 ± 0.2873	10.322 ± 4.138	30.414
	Shared	wCUSUM	0.0269 ± 0.0245	0.6727 ± 0.3222	3.746 ± 1.243	13.858
	Particle	None	0.0405 ± 0.0343	1.2092 ± 0.3909	0.000 ± 0.000	15.017
	Particle	BOCPD	0.0272 ± 0.0224	0.6504 ± 0.2934	9.310 ± 3.972	36.592
	Particle	wCUSUM	0.0276 ± 0.0247	0.6857 ± 0.3302	3.782 ± 1.217	15.969
Mixed	Shared	None	0.0303 ± 0.0220	1.0319 ± 0.2009	0.000 ± 0.000	12.513
	Shared	BOCPD	0.0215 ± 0.0170	0.4714 ± 0.0899	11.867 ± 2.647	30.606
	Shared	wCUSUM	0.0211 ± 0.0148	0.5357 ± 0.0979	3.493 ± 1.005	13.515
	Particle	None	0.0307 ± 0.0222	1.0485 ± 0.2049	0.000 ± 0.000	15.146
	Particle	BOCPD	0.0219 ± 0.0152	0.4959 ± 0.0723	9.893 ± 2.252	39.409
	Particle	wCUSUM	0.0206 ± 0.0134	0.5584 ± 0.1159	3.440 ± 1.056	16.175

Table 8: Diagnostic comparison between B-BRPC-E and a BOCPD-controlled filtering baseline on the synthetic benchmark. Values are mean ± standard deviation across runs. Restart count is reported as a controller-behavior diagnostic. Here restart count denotes hard resets triggered by the BOCPD controller; BOCPD is run at every batch even when no hard reset is triggered.

Scenario	Method	θ-RMSE ↓	y-RMSE ↓	Restarts	Time (s) ↓
Drifting	B-BRPC-E	0.0151 ± 0.0024	0.4919 ± 0.2468	12.16 ± 1.86	11.70 ± 1.57
	BOCPD-WardPFMove	0.0902 ± 0.0302	1.9090 ± 0.1164	0.00 ± 0.00	5.97 ± 0.05
Sudden	B-BRPC-E	0.0212 ± 0.0196	0.6123 ± 0.2934	10.80 ± 4.44	10.18 ± 3.80
	BOCPD-WardPFMove	0.2273 ± 0.2003	1.9929 ± 0.0720	0.30 ± 0.50	5.20 ± 1.94
Mixed	B-BRPC-E	0.0194 ± 0.0167	0.5768 ± 0.2845	11.20 ± 3.90	10.63 ± 3.37
	BOCPD-WardPFMove	0.1870 ± 0.1800	1.9682 ± 0.0948	0.21 ± 0.44	5.43 ± 1.66

J.5 BOCPD-Controlled Filtering Baseline

A natural question is whether BRPC is needed at all, or whether one can simply attach BOCPD-style restart control to a standard filtering or data-assimilation tracker. To test this alternative, we evaluate a BOCPD-controlled moving-particle filtering baseline, denoted BOCPD-WardPFMove. This baseline uses random-walk parameter evolution and joint filtering-style updates, while BOCPD is applied to the resulting predictive evidence.

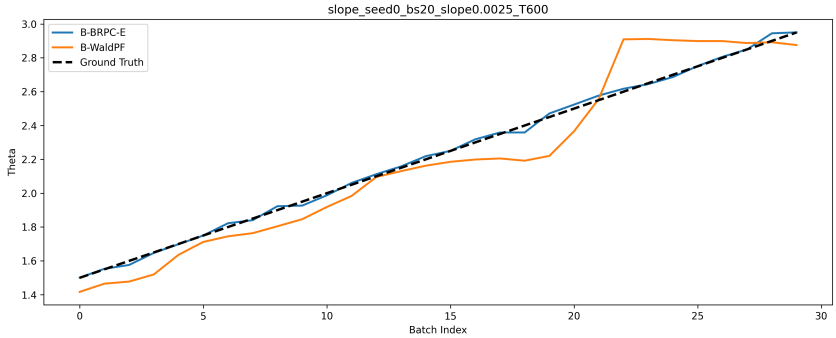
Table 8 compares BOCPD-WardPFMove with B-BRPC-E on the synthetic benchmark. The filtering baseline can move, but it moves through a joint assimilation channel in which parameter motion, simulator discrepancy, observation noise, and regime change are not explicitly separated. Under simulator misspecification, the filtering parameter is therefore asked to explain both the projected calibration target and residual model bias. This creates an online attribution problem: increasing the random-walk variance can make the tracker more responsive, but it also allows the local filter to absorb abrupt changes smoothly and leaves BOCPD with little clean restart evidence. Reducing the variance has the opposite failure mode, making the tracker less responsive after jumps.

The diagnostic reveals an under-segmentation failure of joint assimilation. In the drifting family, BOCPD-WardPFMove triggers no hard restarts on average, yet its θ-RMSE and y-RMSE are substantially worse than B-BRPC-E. In the sudden family, where abrupt changes are annotated, it triggers

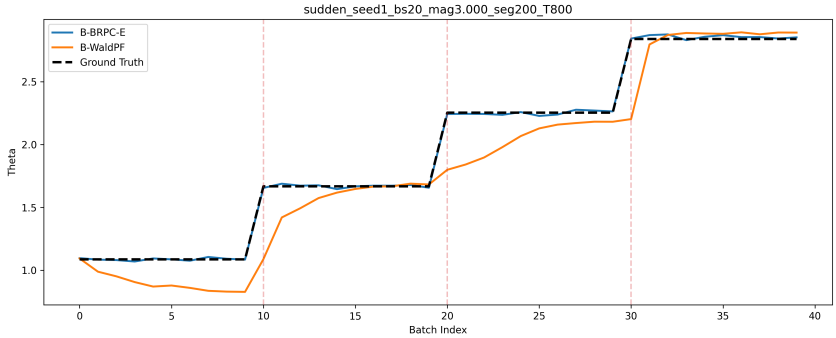
only 0.30 hard restarts on average and has much larger tracking and predictive errors. Thus the filter moves, but it moves through a joint channel that mixes parameter drift, simulator bias, observation noise, and regime change. This smooths away the restart evidence while still failing to recover the projected calibration target accurately.

This behavior is consistent with the online identifiability issue that motivates projected calibration. Appendix F.4 records the static Schur-complement calculation behind this effect. Because the filtering likelihood does not distinguish parameter movement from discrepancy correction, the parameter path can be pulled toward explaining residual model bias rather than only tracking the projected calibration target. BOCPD then sees a smoothed local adaptation process rather than a clean segment-level evidence contrast.

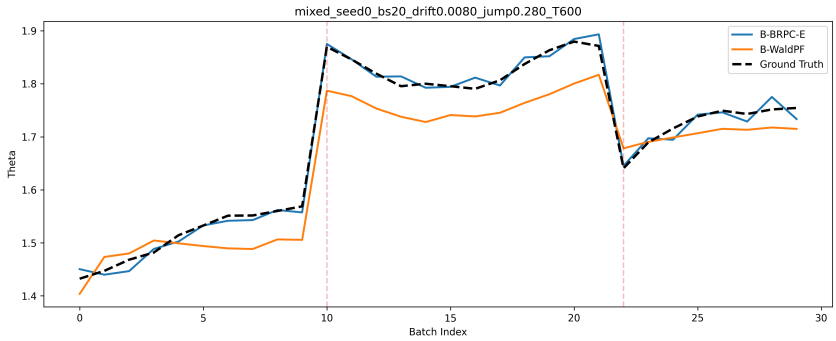
BRPC addresses this issue by separating the two roles. Its parameter channel is projected and discrepancy-free, so particle weights are not directly altered by a flexible discrepancy correction. Discrepancy is then learned conditionally after the simulator anchor is set. This separation does not eliminate all controller–learner interaction, as shown by the over-restart behavior of B-BRPC-E, but it makes the interaction explicit and gives a meaningful object to redesign: either the controller, as in C-BRPC-E, or the BOCPD-facing discrepancy state, as in B-BRPC-RRA. The BOCPD-WardPFMove diagnostic therefore supports the paper's central claim that restart control is not a generic plug-in layer on top of joint assimilation; it must be designed together with the calibration state.



(a) Drifting scenario θ-tracking trajectories



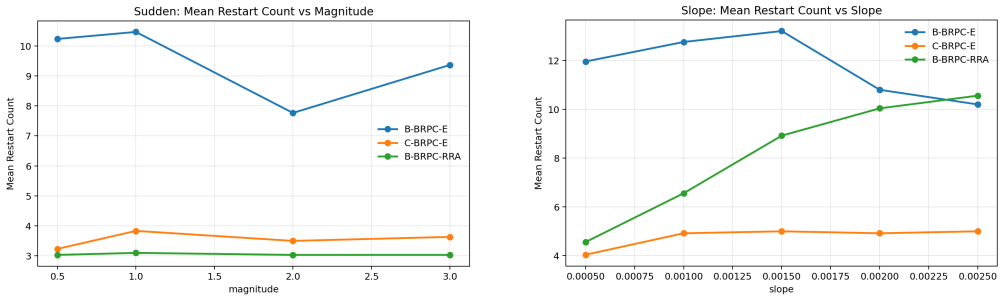
(b) sudden scenario θ-tracking trajectories



(c) Mixed scenario θ-tracking trajectories

Figure 4: Representative θ-tracking trajectories for BOCPD-WardPFMove and B-BRPC-E. BOCPD-WardPFMove can partially follow temporal variation through its random-walk parameter evolution, but its joint assimilation channel blurs the attribution between parameter movement and residual model bias. As a result, it lags or under-reacts after abrupt jumps and provides weak evidence for explicit restart.

J.6 Restart-Count Scaling Diagnostics



(a) Restart count vs. jump magnitude

(b) Restart count vs. drifting slope

Figure 5: Restart count as a function of jump magnitude and drifting slope in the synthetic benchmark. Each point represents the mean restart count over repeated runs for a given scenario configuration.

Figure 5 makes the controller behavior visible beyond the averaged counts in the tables. In the sudden-change panel, **B-BRPC-E** consistently restarts too often, whereas **C-BRPC-E** and **B-BRPC-RRA** both move the restart count back toward the true number of changepoints. Importantly, this is not a story of under-reacting to harder jumps: the improved controllers stay close to the ground-truth level of three restarts across the tested jump magnitudes.

The drifting panel should be interpreted differently. **C-BRPC-E** remains comparatively flat as slope increases, reflecting the fact that its score-based controller is designed to monitor instability in a single pre-update score stream rather than to segment the stream aggressively. By contrast, **B-BRPC-RRA** shows increasing restart frequency as drift becomes faster. In our interpretation, these are not simply false alarms. Because residual re-anchoring recomputes residuals against the current segment anchor and refits the discrepancy model to the accumulated segment data, the outer BOCPD layer also acts as an adaptive memory controller: when the underlying drift is faster, shorter segments and more frequent resets become a reasonable response.

J.7 Controller Hyperparameter Sensitivity

This section checks whether the controller conclusions in the main text are artifacts of a particular BOCPD hazard, hard-restart margin, cooldown, or wCUSUM threshold. We focus on the Exact learner, because B-BRPC-E and C-BRPC-E expose the controller–learner interaction most directly.

For B-BRPC-E, we evaluate four Restart-BOCPD configurations:

$$(\lambda_h, m, c) \in \{(200, 1, 10),\ (400, 2, 20),\ (800, 4, 20),\ (1600, 4, 30)\},$$

where λ_h denotes the hazard scale, m denotes the hard-restart margin, and c denotes the restart cooldown. For C-BRPC-E, we evaluate four wCUSUM configurations:

$$(W, h, \kappa, \sigma_{\min}) \in \{(4, 0.25, 0.25, 0.25),\ (4, 0.50, 0.25, 0.25),\ (8, 0.50, 0.50, 0.50),\ (8, 1.00, 0.50, 0.50)\},$$

where W is the window size, h is the restart threshold, κ is the drift allowance, and $\sigma_{\min}$ is the score standardization floor. All other experimental settings are kept fixed. We report ranges across these hyperparameter grids rather than one row per configuration because the goal is to test qualitative robustness, not to select a scenario-specific optimum.

Event-level Precision@2, Recall@2, F1@2, and Delay@2 are computed only for settings with annotated abrupt changes, namely the sudden and mixed scenario families. A detected restart is counted as correct only if it occurs within two batches after an annotated changepoint; early detections are not rewarded.

Table 9 shows that the main controller conclusion is stable across the tested hyperparameter ranges. B-BRPC-E has high Recall@2 across both sudden and mixed settings, so its main issue is not missing true changes. Instead, its Precision@2 remains low and its restart count remains large across all tested BOCPD configurations. This supports the interpretation that B-BRPC-E over-restarts because ordinary within-regime mismatch is often converted into restart evidence.

Table 9: Controller hyperparameter sensitivity for B-BRPC-E and C-BRPC-E on the representative synthetic benchmark. Each entry reports the range across the tested controller hyperparameter grid listed in the text. Event-level metrics use a two-batch tolerance window and do not reward early detections. Restart count is a controller-behavior diagnostic.

Scenario	Method	Restarts	Precision@2	Recall@2	F1@2	Delay@2
Sudden	B-BRPC-E	9.6–9.8	0.286–0.293	0.933	0.436–0.444	0.10–0.17
	C-BRPC-E	3.0–3.2	0.700–0.867	0.733–0.867	0.714–0.867	0.00
Mixed	B-BRPC-E	12.6–13.0	0.154–0.159	1.000	0.267–0.275	0.20–0.30
	C-BRPC-E	3.2–3.4	0.533–0.600	0.900–1.000	0.667–0.747	0.10–0.30

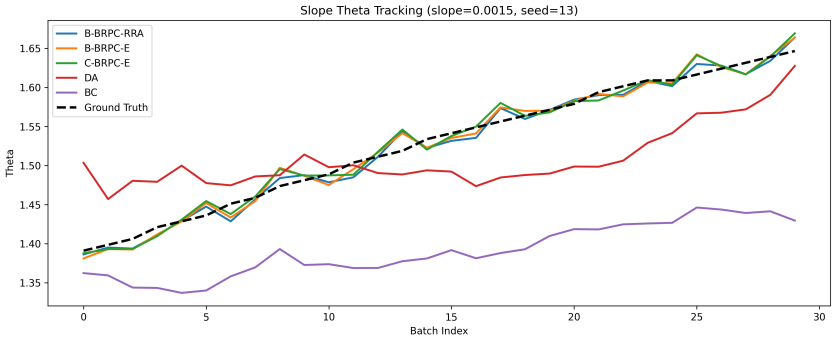
C-BRPC-E shows the complementary behavior. Across the tested wCUSUM settings, it substantially reduces restart count and improves F1@2 in both sudden and mixed settings. Importantly, this is not merely a lower-restart artifact: Recall@2 remains high in the mixed scenario and remains competitive in the sudden scenario, while Precision@2 improves markedly. Thus C-BRPC-E improves event-level restart quality, not only restart quantity.

These sensitivity results also clarify the role of controller choice. BOCPD acts as a multi-expert predictive mixture and hard-restart controller, so reducing its hazard or increasing its margin does not fully remove the evidence-geometry mismatch induced by recursive discrepancy learning. wCUSUM instead monitors a single pre-update score stream and is less dependent on calibrated expert-to-expert likelihood competition. This is why C-BRPC-E is the more robust controller-side intervention, even though it does not necessarily minimize predictive error in every setting.

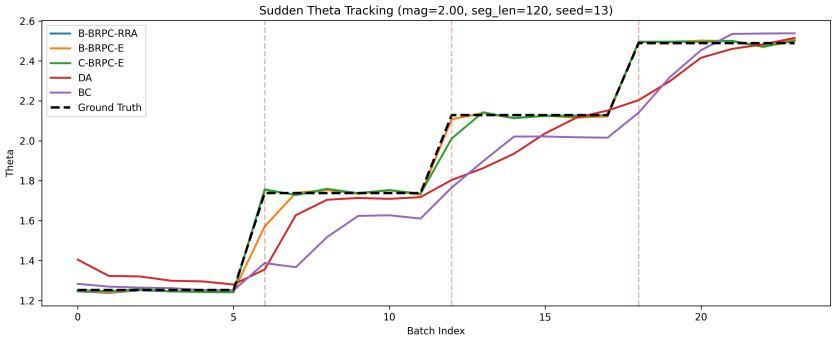
J.8 Representative Parameter-Tracking Trajectories

Figure 6 gives a more direct view of calibration-parameter quality than the aggregate RMSE tables alone. Across the three synthetic scenarios, the figure helps separate two issues that can be conflated in scalar summaries: local tracking quality and restart quality. The Exact learner remains capable of accurate within-regime tracking, but the BOCPD-controlled version tends to introduce more regime fragmentation than necessary. The wCUSUM-controlled version is visually smoother, while residual re-anchoring is most visibly helpful in the jump-driven case, where fast relocation after changepoints matters most.

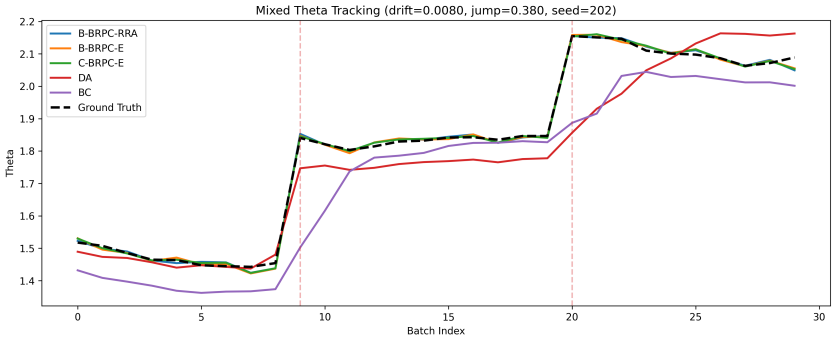
Figure 7 shows that the same qualitative distinctions persist in the plant-simulation benchmark. These trajectories should be read as representative diagnostics rather than as standalone evidence, but they make the aggregate results easier to interpret: **C-BRPC-E** typically preserves the tracking quality of the Exact learner while avoiding some of the excess reset activity of **B-BRPC-E**, whereas **B-BRPC-RRA** is especially convincing in the sudden mode and more mixed in the drifting and mixed modes. Taken together with the tables, these examples support the paper's broader claim that controller design and discrepancy-state design shape different aspects of online calibration quality.



(a) Drifting scenario θ-tracking trajectories

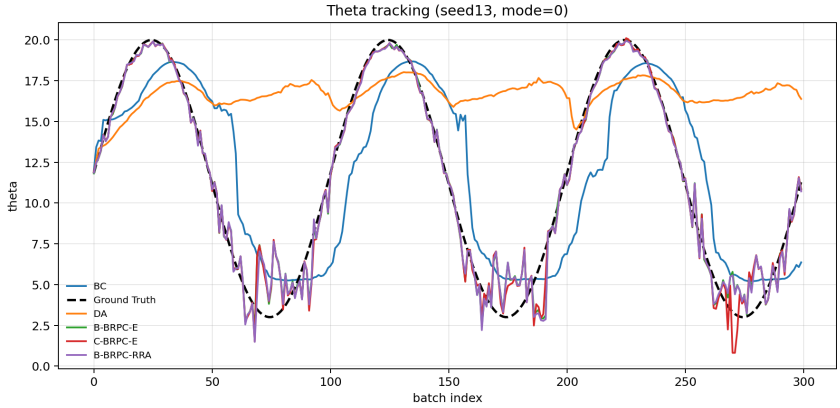


(b) sudden scenario θ-tracking trajectories

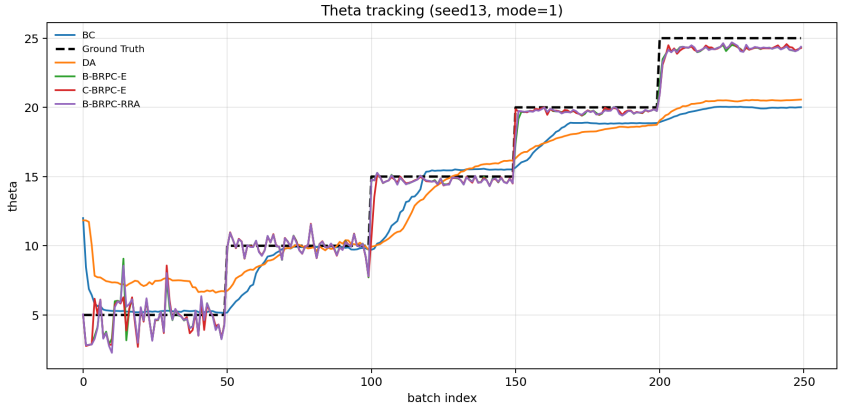


(c) Mixed scenario θ-tracking trajectories

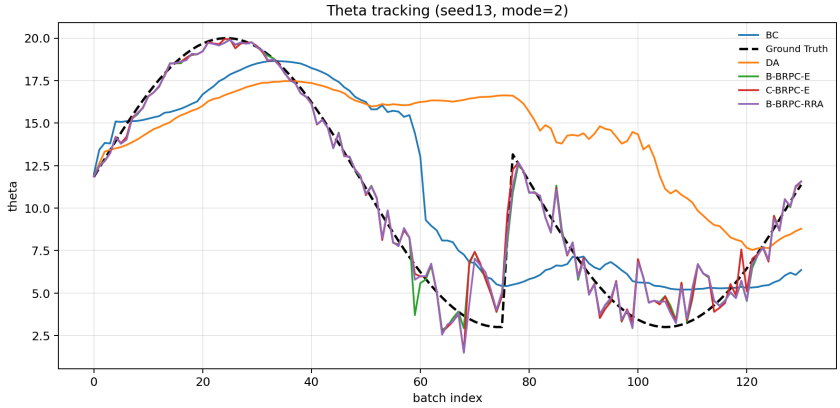
Figure 6: Synthetic Benchmark θ-tracking trajectories for a representative random seed. Each plot shows the true θ trajectory (black) and the posterior mean θ trajectory for each method (colored). The vertical dashed lines indicate the true segment boundaries.



(a) Drifting scenario θ-tracking trajectories



(b) sudden scenario θ-tracking trajectories



(c) Mixed scenario θ-tracking trajectories

Figure 7: Bicycle Plant Simulation θ-tracking trajectories for a representative random seed. Each plot shows the true θ trajectory (black) and the posterior mean θ trajectory for each method (colored). The vertical dashed lines indicate the true segment boundaries.

J.9 Physical-Projected High-Dimensional Diagnostic

This section reports a high-dimensional diagnostic designed to separate three effects: projected-target tracking, point prediction, and the action induced by an outer reset controller. Unlike the main low-dimensional synthetic benchmark, the physical response here is not generated by adding an orthogonal discrepancy to the simulator. Instead, the physical response is generated from a different nonlinear function class, and the calibration target is defined by L2 projection onto the simulator family.

Let $x \in [0,1]^{20}$ and let the simulator have $d_\theta = 5$ calibration parameters. For each time t, the projected calibration target is

$$\theta_t^\dagger = \arg\min_\theta \mathbb{E}_X \left[\zeta_t(X) - y_s(X,\theta)\right]^2,$$

where ζ_t is the physical response and $y_s(\cdot,\theta)$ is the misspecified simulator. In the experiment, $\theta_t^\dagger$ is computed on a large Sobol reference design, and all reported parameter errors are measured against $\theta_t^\dagger$, not against a generative coefficient of the physical function.

We compare four methods. **PF-DA** is a strict Ward-style particle-filter generalization with particles over calibration parameters and kernel length scale. **BC** is a short-window sliding Bayesian calibration baseline with a four-batch window. **BRPC-F** is the fixed-support BRPC learner without outer reset control. **C-BRPC-F** uses the same fixed-support BRPC learner with a hard wCUSUM reset controller. We use $B = 32$, $N = 1024$ particles, $M = 128$ fixed support points, $\lambda_\delta = 1$, and five random seeds.

Physical-projected data generation. The input is $x \in [0,1]^{20}$, and the simulator has $d_\theta = 5$ calibration parameters:

$$y_s(x,\theta) = \sum_{j=1}^{5} \theta_j \phi_j(x),$$

where

$$\phi_j(x) = \sin(2\pi x_j) + 0.5\cos(2\pi x_{j+5}) + 0.25\sin\{2\pi(x_j + x_{j+10})\}.$$

The physical response is generated from a different nonlinear function family:

$$\zeta_t(x) = \sum_{j=1}^{5} a_{t,j} g_j(x) + 0.4h(x),$$

where

$$g_j(x) = \sin(2\pi x_j + 0.4x_{j+5}) + 0.4\cos(2\pi x_{j+10}) + 0.3x_{j+15}x_j.$$

The term $h(x)$ creates locally heterogeneous residual structure:

$$h(x) = \left[1 + 1.5\exp\left(-\frac{\|x_{1:3} - c\|^2}{2s^2}\right)\right] \sum_{\ell=1}^{L} \alpha_\ell \sqrt{2}\cos(\omega_\ell^\top x + b_\ell).$$

Here $c \in [0,1]^3$, $s > 0$, ω_ℓ, b_ℓ, and α_ℓ are fixed across the stream. Observations are generated as

$$Y_{t,k} = \zeta_t(x_{t,k}) + \epsilon_{t,k}, \qquad \epsilon_{t,k} \sim N(0,\sigma^2).$$

The latent coefficient a_t controls the nonstationarity of the physical response, but it is not used as the calibration ground truth. The calibration target is the L2 projected parameter

$$\theta_t^\dagger = \arg\min_\theta \mathbb{E}_X \left[\zeta_t(X) - y_s(X,\theta)\right]^2.$$

Because y_s is linear in θ, this target can be computed on a large reference design. Let $X_{\text{ref}} = \{x_m\}_{m=1}^{M_{\text{ref}}}$ be a Sobol reference design with $M_{\text{ref}} = 50{,}000$, and define

$$\Phi_{mj} = \phi_j(x_m), \qquad \zeta_t^{\text{ref}} = \left(\zeta_t(x_1), \ldots, \zeta_t(x_{M_{\text{ref}}})\right)^\top.$$

We compute

$$\widehat{\theta}_t^\dagger = (\Phi^\top\Phi + \lambda I)^{-1}\Phi^\top \zeta_t^{\text{ref}},$$

with a small ridge $\lambda = 10^{-6}$. All reported θ-RMSE values in this diagnostic are computed relative to $\widehat{\theta}_t^\dagger$, not relative to a_t.

Table 10: Physical-projected high-dimensional diagnostic with $d_x = 20$ and $d_\theta = 5$. Parameter error is measured against the L2 projected target $\theta_t^\dagger$. Values are mean $\pm$ standard deviation across five seeds. F1@2 is reported only for scenarios with annotated discrete changes.

Scenario	Method	$\theta^\dagger$-RMSE ↓	y-RMSE ↓	y-CRPS ↓	Restarts	F1@2 ↑
Drifting	PF-DA	1.0137 ± 0.2174	1.5027 ± 0.0228	0.8834	0.0	–
	BC(4)	0.1845 ± 0.0168	$\mathbf{1.0828 \pm 0.0396}$	0.8066	0.0	–
	BRPC-F	$\mathbf{0.1329 \pm 0.0060}$	1.1213 ± 0.1033	**0.7402**	0.0	–
	C-BRPC-F	0.2038 ± 0.0193	1.1830 ± 0.0264	0.7604	3.8	–
Sudden	PF-DA	1.0506 ± 0.1742	1.5483 ± 0.1604	0.9231	0.0	0.000
	BC(4)	0.2073 ± 0.0297	$\mathbf{1.1223 \pm 0.1157}$	0.8420	0.0	0.000
	BRPC-F	$\mathbf{0.1631 \pm 0.0107}$	1.2173 ± 0.1599	0.7932	0.0	0.000
	C-BRPC-F	0.2030 ± 0.0233	1.1790 ± 0.1512	**0.7426**	3.4	0.303
Mixed	PF-DA	0.9893 ± 0.1943	1.4887 ± 0.0668	0.8781	0.0	0.000
	BC(4)	0.1890 ± 0.0126	$\mathbf{1.0738 \pm 0.0439}$	0.8006	0.0	0.000
	BRPC-F	$\mathbf{0.1555 \pm 0.0254}$	1.2031 ± 0.0378	0.7746	0.0	0.000
	C-BRPC-F	0.1951 ± 0.0395	1.1782 ± 0.0454	**0.7551**	3.8	0.297

Nonstationary physical coefficients. The time variation is applied to the physical coefficient a_t. In the drifting scenario,

$$a_t = a_0 + 0.8\frac{t}{T}v_1 + 0.15\sin\left(\frac{2\pi t}{T}\right)v_2.$$

In the sudden scenario, a_t is piecewise constant with changepoints at $0.25T$, $0.50T$, and $0.75T$, and jump magnitude 1.2. In the mixed scenario,

$$a_t = a_0 + 0.4\frac{t}{T}v_1 + 0.10\sin\left(\frac{2\pi t}{T}\right)v_2 + \sum_{r=1}^{2} u_r \mathbf{1}\{t \geq \tau_r\},$$

with jump magnitude 1.0. The projected target $\widehat{\theta}_t^\dagger$ is recomputed for each batch from the corresponding physical response ζ_t.

Table 10 shows that the fixed-support BRPC learner remains stable in the high-dimensional projected setting. Without outer reset control, BRPC-F achieves the best projected-target tracking in all three scenario families. This indicates that the projected BRPC update itself does not fail in the $d_x = 20, d_\theta = 5$ diagnostic.

The table also separates point prediction from projected calibration. The short-window BC baseline achieves the lowest point y-RMSE, which is expected: with a four-batch window and $B = 32$, BC acts as a strong local refit and can quickly forget old regimes. However, BC is not the best projected-target tracker and has worse predictive CRPS than BRPC-F or C-BRPC-F. Thus the diagnostic distinguishes point prediction from calibrated tracking and probabilistic prediction.

The strict Ward-style PF generalization performs poorly in this setting. This suggests that directly extending continuous-calibration PF to a high-dimensional physical-projected stream is not straightforward. In contrast, BRPC-F retains good projected tracking because the parameter channel remains discrepancy-free and the discrepancy state is learned conditionally.

Finally, the comparison between BRPC-F and C-BRPC-F illustrates the controller–learner compatibility issue in a higher-dimensional setting. The hard wCUSUM controller produces nonzero event-level F1 in the sudden and mixed settings, but it also introduces false alarms in the drifting setting and increases $\theta^\dagger$-RMSE across all scenarios. This behavior is consistent with the evidence-geometry view developed in the main text. In higher-dimensional input spaces, batch-to-batch changes in covariate coverage and residual difficulty can create score fluctuations that resemble regime changes. At the same time, each batch can contain enough information for the local projected learner to adapt without a hard reset. Thus hard reset control is not universally beneficial: in this diagnostic, no-controller BRPC-F is the best projected learner, while C-BRPC-F trades additional variance for some event-level detection ability.